Role of Artificial Intelligence among Small and Medium Enterprise (SME) food stalls in Chennai city

PROJECT REPORT

Submitted to the

UNIVERSITY OF MADRAS

In partial fulfilment of the requirement for the award of the degree of

MASTER OF COMMERCE

Submitted by

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24PCM16

Under the guidance of

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CHENNAI – 600006

March 2026

CERTIFICATE

This is to certify that the project, entitled "Role of Artificial Intelligence among Small and Medium Enterprise (SME) food stalls in Chennai city" submitted to the University of Madras in partial fulfilment of the requirements for the reward of the degree of **Master of Commerce** is a record of original research work done by **Ms. JESHWIN DHIVYA KRUBA J (Register No: 24PCM16)** during the period of 2025 – 2026 of her study in the PG Department of Commerce at Women's Christian College (Autonomous), Chennai- 600006 and the project work has not formed the basis for the award of any Degree or other similar title to any candidate of any university.

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I, **Ms. JESHWIN DHIVYA KRUBA J (Register No: 24PCM16)**, hereby declare that the project entitled **"Role of Artificial Intelligence among Small and Medium Enterprise (SME) food stalls in Chennai city"** submitted to the University of Madras in partial fulfilment of the requirements for the award of Degree of **Master of Commerce** is a record of original research work done by me during 2025-2026 under the supervision and guidance of **Dr. Jenifer Angelen S, M.com., M.Phil., Ph.D.,** Assistant Professor, in PG Department of Commerce, Women's Christian College (Autonomous), Chennai and it has not formed the basis for the award of any Degree or other similar title to any candidate of any university.

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TABLE OF CONTENTS

S.No	Title	Page No.
	Chapter I-Introduction	
1.1	Introduction	1
1.2	Need for study	2
1.3	Scope of the study	2
1.4	Research Questions	3
1.5	Research Objective	3
1.6	Research Methodology	4
	1.6.1 Research Design	4
	1.6.2 Research Instruments	4
	1.6.3 Sampling Technique	5
	1.6.5 Sample Size	5
	1.6.6 Method of data collection	5
	1.6.7 Pilot Study	6
	1.6.8 Statistical Tool	6
1.7	Limitation of the study	7
1.8	Chapterisation	8
	Chapter II – Review of Literature	
2.1	Introduction	9
2.2	Review of Literature	9
2.3	Research Gap	15
	Chapter III – Theoretical Overview	

3.1	Meaning and Introduction of Artificial Intelligence	16
	3.1.2 Concept of AI	18
	3.1.3 Role of Artificial Intelligence	19
	3.1.4 Types of AI	20
	3.1.5 Types of AI tools used in SME food stalls	23
	3.1.6 Merits of Artificial Intelligence in SME food stalls	28
	3.1.7 Demerits of Artificial Intelligence in SME food stalls	30
	3.1.8 Outlook of artificial intelligence in small and medium enterprises (SME) in the future.	34
3.2	Introduction of Small and Medium Enterprise (SME)	
	3.2.1 Introduction	38
	3.2.2 Concept of Small and Medium Enterprises	40
	3.3.3 Role of Small and Medium Enterprises	41
	3.3.4 Evolution of Small and Medium Enterprises	44
	3.3.5 Advantages of SME	47
	3.3.6 Disadvantages of SME	50
	3.3.7 Types of SME	53
	3.3.8 Government support programs for SME in adoptin AI	56
	3.3.9 Key Announcements of Budgets 2026 on SMEs.	61
Chapter IV - Analysis & Interpretation		

4.1	Cronbach's Alpha Reliability Statistics	64
4.2	Descriptive Analysis	64
4.3	Hypothesis Testing	68
Chapter V-Summary		
5.1	Introduction	75
5.2	Summary of Findings	75
5.3	Suggestion	77
5.4	Conclusion	77
5.5	Scope for Future research	78
	Appendix	
	Bibliography	
	Questionnaire	

LIST OF TABLES

S. No	Title	Page No.
1.6.7	Reliability test of pilot study	6
4.1.1	Reliability statistics	64
4.2.1	Demographic Analysis	64
4.3.1	Impact of AI in overall operational performance	68

4.3.2	Awareness and adoption of AI tools	70
4.3.3	Influence of the usage of AI tools and operational efficiency	71
4.3.4	Impact on AI adoption tools and customer satisfaction	72
4.3.5	Challenges and barriers of adopting AI	74

LIST OF HYPOTHESES

Hypothesis No.	Statement	Statistical Test	Page No.
1	There is no significant relationship between the use of AI tools and operational performance of SME food stalls	Regression	68
2	There is no significant difference between the level of awareness and adoption of AI tools among SME food stall owners.	ONE WAY ANOVA	69
3	There is no significant difference between the influence on usage of AI tools and the operational efficiency.	Regression	70
4	There is no significant relationship between the adoption of AI tools and customer satisfaction	Regression	71
5	There is no significant difference in the level of challenges faced by SME food stalls on basis of adopting AI.	ONE WAY ANOVA	73

INTRODUCTION

1.1 Introduction

The food stall and small business sector plays a major and important role in Chennai City. The Small and Medium Enterprise (SME) food stalls such as street food stalls, small restaurants, cloud kitchen, bakery, food truck and beverages which results in fast and affordable food services and also increase employment opportunity. In the recent times, digital technologies and Artificial Intelligence (AI) based tools have slowly entered this food sector through billing software, stock and inventory applications, order management systems and digital payment platforms.

Artificial Intelligence helps in performing duties and tasks like forecasting future demand, inventory control, smoother services, easy and convenient Billing as well as the takeaway outlets. These technologies have developed and improved operational efficiency, minimize human mistakes, reduce food wastage and improve consumer services and meet their needs on time. During rush peak hours where it is unable to provide customers needs on time and some orders can be missed out therefore AI helps the food stall owners by manage their stock and inventory where provides information such as out of stock which also prevents the wastage of food as well as meet customer needs.

Even though the increasing of digital and AI based applications and technologies such as Point of Sale (POS), mobile billing such as Phone pay or Google pay and delivery platforms dashboards such as Swiggy or Zomato, many SME food stall owners in Chennai are still using and rely on traditional or manual practices. The level of awareness, understanding and adoption of AI technologies varies greatly among food stall operators. The major challenges such as high cost, poor digital knowledge as well as the infrastructure and fear of using Artificial Intelligence tools are considered to be the challenges and barriers in adopting AI. Particularly, this study may not fully focused on every business activities and operations other than billing, inventory management, delivery services and customer handling.

Previous studies and research articles proposed, The Small and Medium Enterprises in adoption of digital and technology has helped their business growth, where their works becomes faster and easier and also the customer are satisfied and happy with their services

However, limited empirical research has been done on the role of Artificial Intelligence among Small and Medium Enterprises food stalls in Chennai City.

This study focuses on understanding the role and impact of Artificial Intelligence in developing the overall day to day performance of Small and Medium food stalls in Chennai City. The study tries to find out how Artificial Intelligence tools provides the billing, stock and inventory control and management, smooth delivery services and customer handling. Moreover the study also identify the challenges and barriers faced by Small and Medium Enterprises food stall owners in difficulty in adopting Artificial Intelligence.

1.2 Need for the study

The sudden growth of digital technology and AI has created new opportunities for SME food businesses to improve their productivity and competitiveness. And even so, many Small ad Medium Enterprise food stall owners are still unaware of the possible advantages of Artificial Intelligence and also digital technology. It is necessary for SME food stall owners to understand the needs of AI based tools to support their day-to-day operations such as billing, stock and inventory control management, smooth delivery services and also customer satisfaction.

Understanding the level of awareness and adoption of Artificial Intelligence among SME food stall owners the Government, technology providers and local business associations help them to provide training and support programs. The study is also need to understand the practical challenges and barriers faced by Small and Medium enterprises while using AI tools such as high cost, lack of skill and technical knowledge. Therefore, this study is important to know the real role of Artificial Intelligence in developing the operational performance and long term survival of SME food stall owners in Chennai City.

1.3 Scope of the study

The present study of Small and Medium Enterprise food stall is limited particularly in Chennai City. The study focuses especially on the role of Artificial Intelligence and digital tools in day-to-day operational activities such as billing, stock management, smooth delivery services and customer satisfaction.

The study does not involve big restaurants, hotel chains, franchise outlet and corporate food companies. It focuses only to small and medium food stalls such as Bakeries, Cloud Kitchen, Beverages, Small restaurants, Takeaway outlets. The

geographical scope of the study is focused only on Chennai City and therefore the results of other cities or regions is not applicable.

1.4 Research Questions

The present study pursue to answer the following research questions:

1. What is the level of awareness and adoption of Artificial Intelligence based tools among SME food stall owners in Chennai City?
2. How do AI tools controls the operational efficiency in billing, inventory management and delivery services?
3. How does the use of AI-based technologies influence customer satisfaction and their involvement in SME food stalls?
4. What are important challenges and barriers faced by SME food stalls in adopting Artificial Intelligence tools and technologies?

1.5 RESEARCH OBJECTIVIES

Primary Objective

- To assess the role and impact of Artificial Intelligence in enhancing the overall operational performance of Small and Medium Enterprises in Chennai City.

Secondary Objective

- To identify the level of awareness and adoption of Artificial Intelligence based tools among Small and Medium Enterprises food stall owners in Chennai city.
- To evaluate the effect of Artificial Intelligence tools on operational efficiency and also billing, inventory management and delivery services.
- To analyse the role and effect of Artificial Intelligence in developing customer satisfaction and their involvement in Small and Medium Enterprise food stalls.
- To examine the important challenges and barriers confronted by Small and Medium Enterprise food stall owners in adopting Artificial Intelligence tools and technology.

1.6 Research Methodology

1.6.1 Research design

This study adopts descriptive and analytical research design using quantitative approach to examine the role and impact of Artificial Intelligence among SME food stall owners. The descriptive research design is used to describe the current condition of awareness and adoption of AI-based tools as well as the background details of the respondents such as type of food stalls, years of business experiences, nature of

performance and use of digital devices among SME food stall owners through a structured questionnaire. This study also adopts the analytical research design as it examines the identity of the AI usage and day-to-day performances such as billing, inventory management, delivery services and customer satisfaction. And the study also follows a cross-sectional design where the data are collected from the respondents at a single point of time in Chennai city.

1.6.2 Research Instruments

The primary research instrument for data collection is a structured questionnaire. The questionnaire consists two parts, the first part is demographic profile of the SME food stall owners related to their age, educational qualification, type of food stalls, nature of ownership, years of business experience, etc., and the second part consists of awareness and adoption of AI, AI and operational efficiency, AI and customer satisfaction, impact on overall performance and challenges and barriers in adopting AI. All the statements are measured using a 5-point Likert scale, ranging from 1-strongly disagree, 2- disagree, 3-neutral, 4-agree, 5-strongly agree.

1.6.3 Sample

In this study, the sample consists of the Small and Medium Enterprise owners and managers in among Chennai City. The respondents which include owners from different categories of food stalls such as street food, small restaurants, cloud kitchen, bakery, food truck and beverages. The respondents were from different locations in Chennai City.

1.6.4 Sampling Technique

This study uses the non-probability convenience sampling techniques for collecting the data. As the owners or managers of SME food stall are selected on the availability and willingness in the survey. This sampling technique was termed and considered to be appropriate due to limited time and it was not possible to collect from all SME food stalls in Chennai City.

1.6.5 Sample Size

The sample size consists of 200 respondents belonging to the SME food stall owners of adopting AI in Chennai City. The selected sample size is considered appropriate as it gives enough responses from different types of SME food stalls such as street food stalls, small restaurants, takeaway stall, bakery, food truck and beverages. The sample size also gives enough data to understand the level of awareness, adoption and impact of AI based tools in overall performances of SME food stalls in Chennai City.

1.6.6 Method of data collection

The data are collected on both primary and secondary data source.

Primary Data

For the purpose of collecting primary data through primary source, a structured questionnaire was designed and distributed among the SME food stall owners in Chennai City. The questionnaire consists two parts, the demographic section which has age, educational quantification, type of food stalls, nature of ownership, years of experience, etc., and the second part consists of awareness and adoption of AI, AI and operational efficiency, AI and customer satisfaction, impact of overall performance and challenges and barriers in adopting AI. All the statements are measured using a 5-point Likert scale, ranging from 1-strongly disagree, 2-disagree, 3-neutral, 4-agree, 5-strongly agree. The data were collected in both online and offline mode.

Secondary Data

The secondary data were collected from various journals, published articles websites related to SME food stall adopting Artificial Intelligence and also their challenges and barriers. These sources helps to know about the level of awareness, adoption and impact of AI based tools in overall performances of SME food stalls in Chennai City.

1.6.7 Pilot Study

Pilot study is a initial study taken to test the questionnaire design and collection of data before the actual full data is collected. This study is conducted to test the questionnaire sustainability, clarity and productivity of the research questionnaire and collection of data. The main aim of the pilot study is to test the quality of questionnaire in order to get a accurate results in the final study.

Table 1.6.7.1

Reliability Test

Cronbach's Alpha	No of items
0.888	25

(Source: Primary data)

This table shows the Cronbach's Alpha value for all the questions using SPSS. Cronbach's Alpha is a statistical tool that is used to examine the reliability and internal consistency which shows all the set of items in the questionnaire are closely related to each other. The Cronbach's Alpha score in the above table is 0.7 which is good and accepted.

1.6.8 Statistical Tools

This study uses the Cronbach's Alpha to study the reliability and internal consistency of the questionnaire. Percentage analysis was used to determine the demographic profile of the SME food stall respondents. Descriptive analysis is collected from the awareness and adoption, overall performance, operational efficiency, customer satisfaction and challenges and barriers of AI.

1.7 Limitation of the study

The study is undertaken only among the SME food stalls in Chennai City, therefore the findings does not used to food stalls in other places rather than in Chennai City. This study may not be fully correct as it use information by a particular and limited number of shop owners. The study focuses only on particular business

operations and activities such as billing, inventory management, delivery services and customer handling and does not include other. More over, the data were collected only one time due to limited time period and the use of AI tools over a longer period could not be studied.

1.8 Chapterisation

Chapter 1

This chapter includes the Introduction to the study, Need for the study, Scope of the study, Research questions, Research methodology, Limitations of the study and Chapterisation.

Chapter 2

This chapter includes the Literature study and Identification of research gap.

Chapter 3

This chapter includes the Conceptual framework, definition and terminologies, relevant theories and models.

Chapter 4

This chapter includes the statistical analysis and Interpretation of results.

Chapter 5

This chapter includes the Major findings, Suggestions and recommendations, Conclusions and Scope of future research.

REVIEW OF LITERATURE

2.1 Introduction

A review of literature is a systematic review of past literature related to specific field of research. It gives a proper idea about the study of the previous researchers and helps in finding their key concept, variables and their findings and results. The main purpose of the review of literature is to analyse the previous study and help in identifying the research gaps, unsolved findings and the areas where the research topic can be further done. The review of literature helps to remove the overlapping of the present study from the previous study.

2.2 Review of Literature

1. Aljohani et al. (2025) conducted a study on Artificial Intelligence adoption and strategic decision-making: Study of Small and Medium Enterprise in Tokyo. The study aims the adoption of AI that affects the decision making, operational efficiency and competitiveness in Tokyo. The data were collected from journals and reports using a systematic literature review. The findings says that the 23% of SME in Tokyo have adopted AI in manufacturing and retail sectors. The barriers includes high costs and limited skills which improves overall performances.
2. Kalleparambil (2025) conducted a study on AI organizational readiness for SME. The study is based on improving a conceptual model in SME organization, strategic and technological ecosystem the adoption of AI in SME. The study uses qualitative approach and the findings says that proper government training, limited resources, lack of technology are the challenges.
3. Rahmani et al. (2025) conducted a a study on Adoption of Artificial Intelligence (AI) in Small and Medium Enterprise E-commerce to improve marketing performance. This study aims at understanding the factors influencing AI adoption aiming e-commerce SME's in Indonesia and their performance in the market by Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). The data was collected from 155 e-commerce SME's in Jakarta using a structured online survey. The findings says that the PU works better than PEOU and also using AI makes better marketing and improve marketing performance.
4. Sultan et al. (2025) conducted a study on Market influence to use AI on the operational performance of Small scaled food restaurants. This study aims on AI helps the day to day performance in SME restaurants using Technology

Acceptance Modal (TAM) to easily understand and accept AI in SME. The data were collected from 100 SME owners in Pakistan through a survey questionnaire using quota sampling. The findings says that using AI improves and increase productivity and day to day performance smoothly.

5. Indiran et al. (2025) conducted a study on Artificial Intelligence adoption in Malaysian SME: When to optimize and when to transform business models. The study aims Optimization of resources or business transformation of adopting AI of different sectors in Malaysia. The data were collected from 10 SME owners from home-based food services using in-depth qualitative interviews. The findings says that Malaysian SME use AI to make overall operation performance easier such as creating content, answering the customer queries and automated schedules.
6. Jagannathan (2025) conducted a study on Transforming the future strategic AI adoption for small food and beverage business. The study focused on small food business use and adoption of AI in better and legal way. The data are collected from 9 small food and beverages business in Canada using surveys and semi-structured interviews and the questionnaire was prepared and based on Technology Acceptance Model (TAM). The findings says that the business owners and managers see AI has an helping tool for growth in marketing and customer services but the hindrance and the challenges is lack of skill, fear of using AI and high cost.
7. Mohib et al. (2025) conducted a study on the challenges and limitations of Artificial Intelligence adoption in Small and Medium-sized Enterprises. This study aims at understanding the challenges which is important to enhance and encourage technology development. Government officials, AI solution vendors and SME management must understand the reality usage of AI and their benefits. From this purpose the data was collected from 150 SME from different sectors using a structured questionnaire. The findings says that the SME struggle with cost, skills and data but they still believe Ai is uesful.
8. Moorthy et al. (2025) conducted a study on the impact of Artificial Intelligence on the growth and performance of Small and Medium Enterprise with reference to increase industries in Coimbatore. This study aims at find out how much SME knows about AI and how AI helps their business grow. The data is collected by using collected by using convenient sampling method by selecting 60 SME by the

structured questionnaire. The findings says that AI improves the overall performance of SME by increasing efficiency, reducing costs, enhancing decision making.

9. Visuthipol et al. (2025) conducted a study on Artificial Intelligence-enabled decision making in social media adoption for sustainable digital business in Thai SME. This study looks how SME use social media and how it helps their business grow and become more sustainable in the long run. The data is collected by using structured questionnaire from 776 SME owners through online. The findings states that when SME use AI they build enterpreneurship.
10. Makinde (2025) conducted a study on Enhancing SME performance through AI in Africa. This study aims to examine how technology affects SME performance. The data are collected from 436 owners managers of manufacturing SME and used survey method. The findings says that through AI in SME it improves the performances of SME.
11. Lambert et al. (2025) conducted a study on Entrepreneurship of SME emerging AI in Russia. This study aims to SME performance using AI is better. The data is collected from 47 SME firms and used cross sectional survey. The findings says that the AI governance, ethics and employee skills are more important than the high AI usage.
12. Bhalla (2025) conducted a study on the role of AI in MSME and its impact on financial performance and business sustainability. This study aims to improve financial performance and SME sustainability. The data are collected from MSME business and the technique used is qualitative case study analysis. The findings says that AI improves the inventory control and fraud detection and improves financial performance and sustainable business growth.
13. Alwakid et Dahri (2025) conducted a study on AI capabilities and green entrepreneurial orientation for sustainable SME performance in Saudi Arabia. The data are collected from 250 SME managers and used survey method with Structural Equation Modeling(SEM). The findings says that using AI improves sustainability and green innovation.
14. Nguyen et al. (2025) conducted a study on AI assisted big data analytics capabilities, knowledge and innovation of performance in SME of Vietnam. This study aims to examine AI-analytics by digital knowledge in capability and improves innovation.The data are collected from 292 manufacturing SME and

mixed survey methods were used.The findings say that AI improves innovation and digital knowledge.

15. Zavodna et al. (2024) conducted a study on Barriers to implementation of Artificial Intelligence in Small and Medium-sized Enterprises. The paper's main goal is to help organizations by understanding the barriers of adopting AI so they can create customized solutions, provide guidance and make AI easier and more effective. The data was collected form 22 interviews. The findings says that the AI adoption is slowed down by low trust, limited knowledge, weal technology and a shortage of skilled AI professionals.
16. Hewage et al. (2024) conducted a study on Artificial Intelligence adoption on customer engagement: A qualitative study on fast-food SME. The study aims the factors which influence the non-adaption of AI for customer engagement in Sweden using UTAUT framework. The data were collected from fast food SME owners through qualitative interviews. The findings says that the lack of customer engagement in using AI tools, lack of skill and resources are the factors which influence the non-adaption of AI.
17. Jia (2024) conducted a study on The Impact of Artificial Intelligence and big data analytic on Supply Chain Performance of SME in Chinese fresh food E-commerce Industry. The study says AI improves the supply chain management in SME in China especially after COVID-19. The data were collected from SME owners of fresh food e-commerce sector in china through qualitative methods by in interviews and observation. The findings says that the cost challenges in supply chain management, high risk and uncertainties and wastage in their food business and AI helps to improve the overall performance.
18. Hussain et Rizwan. (2024) conducted a study on Strategic AI adoption in SME. This study aims on adoption of AI in SME and the frameworks like high cost, lack of skill and technology, owner acceptance and lack of training to employees. The technique is used is conceptual and framework. The findings says that a step by step process helps the efficiency of SME.
19. Iyelolu (2024) conducted a study on Adoption barriers and future growth opportunities of AI solutions in SME. This study aims on the benefits, challenges and planning for AI adoption in SME. The technique used here is literature review approach. The findings says that the AI improves the productivity,

innovation and competitiveness. The high cost and lack of skills are the major barriers.

20. Kandeel et al. (2024) conducted a study on the Impact of AI applications on the performance of SME. This study aims to examine recent AI developments improve SME productivity, profitability and employee efficiency. The findings says that AI increases SME performances and support business growth.
21. Ahmed (2024) Conducted a study on Artificial Intelligence (AI) Technology Adoption in SME. This study aims at the Malaysian SME find difficult in adopting AI through policy makers that is the Government, evidences collected from SME of Malaysia and AI based innovations. The data is collected from interviews with SME owners and about four companies were examined with a structured questionnaire based on Technology Organization Environment (TOE) framework. The findings says that the high cost of AI tools, lack of understanding, low organizational adoption and uncertainty risks are the major challenges.
22. Arroyabe et al. (2024) conducted a study on Analyzing AI adoption in European SME's: A study of digital capabilities, innovations and external environment. This study aims on digital abilities and business support which affects the adoption of AI in SME using two theories such as dynamic capabilities and resource dependency. The data were collected from 12,108 SME in the EU using regression analysis.The findings says that the digital capabilities and innovation capabilities are combined with strong internal capabilities which is more important than the external support for adoption of AI.
23. Cooper(2024) conducted a study on SME use of AI for new product development (NDP) in Ireland.The data are collected from 47 Irish SME and the technique used is survey method and descriptive quantitative analysis.The findings says that AI adoption and readiness are low due to high cost.
24. Gao et Segumpan (2024)conducted a study on the effect of A driven talent management on organizational performance in retail SME.This study is based on AI recruitment, training and performance appraisal influence retail SME performance and employee engagement.The technique used is systematic literature review.The findings say that AI improves organizational performance in retail SME especially on HR practices and higher employee engagement.
25. Schonberger (2023) conducted a study on Artificial Intelligence for Small and Medium Enterprises: Identifying key applications and challenges. This study

looks how SME in Germany are suing AI and investigate the current state of using AI and guides the industry leaders to support and adopt AI. The data is collected by using standardized online questionnaire. The findings says that SME in Germany has many challenges like high costs, skill shortage and improper training support which leads a lack of successful in future.

26. Lada et al. (2023) conducted a study on Determining factors related to Artificial Intelligence (AI) adoption among Malaysia Small and Medium-Sized businesses. The study aims to find out which factors help or hinder SME from adopting AI usage in different industries in Sabah, Malaysia. The data is collected by using survey online questionnaire from 196 respondents. The findings says that SME can successfully adopt AI when strong leadership commitment and improving internal readiness.
27. Bhalerao et al. (2022) conducted a study n Barriers and benefits of AI adoption in Small and Medium Enterprise. This study aims at adopting AI and how AI can benefit the by improving decision making, employee management, customer development, inventory control and cyber security. This paper is based on literature review by collecting data from various sources. The findings says that small business finds hard to use AI where it helps them to do their work easily and better.
28. Liu et al. (2021) conducted a study on smart stall Artificial Intelligence solution vendor economy and its transformation model from the perspective of Artificial Intelligence. This study aims at AI which can be integrated into street stalls during COVID 19 to improve market efficiency towards 3.0 evolution. The study is based on descriptive and conceptual which focuses on smart stalls and their transformation under COVID 19. The findings says that AI helps street stalls more flexible, efficient and low cost.
29. Basri (2020) conducted a study on Impacted of AI assisted in social media marketing on performance of SME in Saudi Arabia. This study aims to examine the impact of AISMM on the performance of start up SME. The data are collected from start up business of SME using Survey method. The findings says that AI increases customers, improves business profit.
30. Ardito et al. (2020) conducted a study on AI adoption and income growth in European SME and its synergies. This study aims to examine AI adoption increases the revenue growth and strengthen IoT and BDA. The data are collected

from 11,429 European SME and the technique used is quantitative secondary data analysis. The findings says that AI adoption increases SME revenue and growth.

2.3 Research Gap

Previous study has stated the Adoption of AI on among SME food stalls in various countries but there is no particular study on the adoption of AI on among SME food stalls in Chennai City. There are very few studies on SME food stalls particularity on the operational performance and the awareness and adoption of Artificial Intelligence based tools.

Theoretical Overview

3.1 Meaning and Introduction of Artificial Intelligence

Artificial Intelligence (AI) is latest modern technology which helps machines, computers and electronic devices think fast as well as in a smartest way. It makes the complicated works by breaking them into simple words by learning past data and give simple decision making. Particularly in business, AI helps to make their overall operational work easier, faster and also accurate without any mistakes. For example, AI can help in the preparation of bill because nowadays GST is termed to be crucial role, Managing stock and inventory, examining the sales and also understand the customer behaviour. Nowadays, AI is slowly adopting into small business like SME especially in food and retail sectors.

The Small and Medium Enterprises (SME) are the small business which has limited capital, a small business units and structures and a small number of employees. In SME, there will be regular and daily sales and customers. The business is operated under traditional methods and only the owners and managers will operate their whole business structure by themselves. Mostly, every SME use manual records, simple mobile applications or very basic digital payments systems like Phone pay or G pay to record the sales and incomes.

In the present study, the focus is mainly on the SME food stalls operating in Chennai City. Chennai is the biggest metropolitan city and also a major urban cities in India. It has more and large number of SME food stalls in every nook and corner of Chennai like markets, bus stops, busy streets, office areas and residential locations. SME plays a vital role in Indian economy by increasing employment opportunities and encouraging small entrepreneurship.

The background of SME food stalls in Chennai city follows the traditional and experience based business practices such as manual billing through hand or simple and basic mobile applications, stock and inventory is observed by the manager itself. The other activities such as planning, pricing and purchase of raw material is done by past experience and their own point of view. It might look simpler but it might lead billing errors, wastages, low stock and over purchasing.

The present state of SME changes over the past years. Customer get faster and better services. The digital payments such as Gpay and Phonepay plays a vital role. At present, there are many food stalls in Chennai and the competition among them is also

increases. Therefore, cost like labour, fuel, transportation, raw material increases due to the increase of competition among the food stalls in Chennai. This is a major hindrances to maintain the overall performances of their business.

Nowadays the necessity of technology is became vital. Digital technology and Artificial Intelligence helps the food stall owners to manage the overall performance such as billing, sales, stock control and reduce mistakes. AI also provides advance support such as assessing past sales data, forecasting future demand and advising better business decisions.

There are some AI tools which supports the inventory management by giving notifications about the present level of stock such as “Out of stock” or “Refill the stock” based on the consumption patterns of the customers. This helps mainly in the peak hour customer time where its hard to manage each and every thing and the main goal of the business is to meet the customer needs.

Another significant of AI is meeting the customer interactions and satisfaction. Customer is the king of business and the competition is very high at present. Its important to study the customer minds and their convenience. Here, AI collect customer feedback and complaints analyse them and gives improvements. Sometimes AI gives offer suggestions to regular customers where these activities helps to gain customer relationships and encourage them to visit their food stall again.

Artificial Intelligence is becoming common place in the business world where it is automated to handle the routine tasks and enhance efficiency. The AI based tools are employed in the process of inventory management, customer preferences analysis, and sales trends forecasting by many companies.

Technologies can assist the businesses to cut the cost of operations, enhance productivity and better decisions made in strategic matters. The SME are also slowly embracing the use of AI tools on their day to day life of operations. The conventional ways of conducting business are usually based on the manual keeping that can be rather time consuming and subject to mistakes.

To conclude, AI is a powerful technology which helps human work more easier and effectiveness without any errors.

3.1.2 Concept of AI

Artificial Intelligence (AI) is the ability of a machine to be able to behave like humans, including the resolution of issues, learning, and decision making. With data

processing and pattern recognition, the AI systems can analyze large amounts of data in a short period and gives insights that aids in making informed decisions by the businesses. This processing and interpretation of information in a high rate enables organizations to work more effectively and tackle issues more accurately.

Some of the technologies under AI include machine learning, natural language processing, and computer vision. Machine learning, through its ability to learn using historical data, can become part of the system to enhance its performance, whereas natural language processing can enable machines to comprehend and communicate in human language. Computer vision also provides systems with the capability to identify and interpret the visual information, which has proved to be very handy in quality control, monitoring and automation of operations. These technologies combine to ensure that AI becomes an effective method of streamlining processes and incorporating intelligence to the routine business operations.

The fact that AI can minimize the necessity of human intervention is one of the most critical advantages of AI. Automation of repetitive and routine tasks can reduce errors and help to save time among the employees. This improves precision in such spheres as billing, stock operation, and reporting, which leads to a higher level of reliability in operations and enables human workers to invest more time in more significant or creative activities. AI also offers predictive data, which allows companies to establish trends in the future considering previous information and allocate resources.

AI is very scale-able and flexible and is applicable to small as well as big businesses. Large-scale AI solutions can be implemented only at the growth of small businesses, which can start with the installation of simple AI solutions (automated billing or inventory management systems, etc.). Moreover, AI could deliver personalized experience because it analyzes customer preferences, recommends customized services, and assists in pursuing an individual approach to marketing. AI also helps to minimize manual labor, wastage, and delays during operation and make the businesses more cost-effective and competitive.

3.1.3 Role of Artificial Intelligence

AI is vital in enhancing performance and efficiency of SME food stalls. Among the greatest effects is the effect on fast billing and order processing. Point of sale (POS) systems with AI capabilities have the capability of processing multiple orders in real time, minimize waiting time, and achieve correct billing. This does not only enhance customer satisfaction but also enables the stall owners to deal better with peak periods of the day which ensures that they operate smoothly all through the day.

Another aspect that AI is invaluable is inventory management. AI prevents shortages and overstocking by predicting the needs of a particular stock and tracking its real-time status. This makes sure that popular ones are constantly in stock and wastage is kept to a minimum. Automated inventory control also enables stall owners to plan the procurement processes better and decrease the costs that should not be incurred and increase the overall resource management.

The AI also increases the efficiency of the operations by eliminating human errors. Automated computations, tracking and reporting reduce errors in billing, handling stock and daily operations. Such common operations as report creation, sales records upkeep, and inventory monitoring can be fully automated, which saves time and gives stall owners an opportunity to work with customers, create new menu items, and develop the business.

AI also improves customer experience. Customers can be analyzed in their preferences, buying history, and their feedback, and the systems will give a personal offer and recommendations. AI also has a useful operational decision support and assists the owner in the planning of menu changes and promotions and pricing strategies. Providing information about daily business choices and allowing the scaling of the business, AI allows small stalls to develop their services effectively, cost-efficiently, and have a competitive edge in the saturated market.

3.1.4 Types of AI

1. **Rule-Based AI Systems**

- The rule-based AI systems perform tasks strictly in accordance with the instructions and rules that have been pre-programmed in them and do not take previous experience into account or get adjusted to new patterns.
- They automatically add the total amount of the customer orders, all the menu items ordered and are able to bill the orders accurately and quickly.
- The system uses taxes, service fees and discounts in a similar manner and the rules on which the business operates, eliminating any possibility of manual errors.
- AI-based on rules will be able to produce sales reports that include the total revenue, sales of an item, and other indicators in specific amounts based on the registered transactions on a daily or weekly basis.
- It will change the pricing of items that are being promoted, provided as a combo, or a special deal precisely based on the programmed parameters.
- To keep business records organized, transactions are automatically captured in a standard form which consists of time, items sold as well as the total amounts.
- The system will be able to alert when inventory levels drop to a predetermined minimum level to ensure that replenishment of stock is done according to predetermined levels.

2. Machine or learning based AI

- AI systems based on machine learning perform the analysis of past data with the help of algorithms, and such systems are able to identify tendencies in sales, customer preference, and inventory utilization and become better predictors as time goes on.
- Such systems will be able to determine the dishes that sell well during which days or times, such that the stall owners will prepare the foods more effectively.
- They are able to predict the future demand of customers using the history of the sales and therefore planning is better in terms of inventory and shortages are reduced.
- The machine learning AI can suggest what products should be replenished more regularly, which involves buying habits during weeks or months.

- They will be able to identify seasonal or weekly changes in customer preferences and thus menu planning and promotional activities will be based on the demand.
- These systems give an understanding of customer purchasing patterns and this will assist the stall owners to determine what to include in a menu, pricing tactics and marketing initiatives.
- With time, machine learning AI will be more precise in trend prediction, which allows small business owners to make evidence-based operational choices.

3. Artificial Intelligence on Data Analysis and Forecasting.

- The information that is processed in data analysis and forecasting AI tools includes business information e.g. daily sales, inventory, and customer purchase history to give actionable insights.
- They are able to produce detailed reports of the performance of individual menu items, total revenue earned per day, and the sales trend of the overall sales during a certain period of time.
- These tools know the times of the day and weeks when there is the greatest demand of customers and hence enable owners to plan their staff and prepare ingredients optimally.
- Predicting characteristics forecast based on past information the future sales volumes, and can assist the owner of the stall against stockpiling or even out of stock.
- They are able to point out items or product categories, which are performing poorly in order to direct owners to change their menus or promotional offers in order to drive sales.
- These AI applications can be used to make strategic decisions by providing a clear understanding of the revenue trends, customer preferences, and inventory turnover.
- Through the trend analysis over the years, such systems help the stall owners to make better plans on promotions, pricing and procurement, which makes the business more efficient.

4. The Customer Interaction AI Tools.

- The AI-based systems of customer interaction process and store the data on the customer preferences, the feedback, and purchasing patterns to enhance the quality and customer satisfaction.
- The system of digital feedback allows customers to score their experience and leave comments about the quality of food, service, and other elements of the stall.
- The chat-bots or automated messaging services can address customer inquires, process orders, and give information on the menu-item or promotions.
- The AI will be able to process the behavior of customers and define what is most popular and what may need improvement or improvement.
- The integration of CRM enables storing customer information and providing them with personalized promotions, loyalty points, or even discounts on regular customers.
- The negative feedback may be notified automatically and the stall owners may react promptly and take corrective actions to maintain customer satisfaction.
- With the ability to see customer trends and tastes, such systems are useful to enhance service, enhance retention, and direct product or menu changes

5. AI POS and Billing Systems

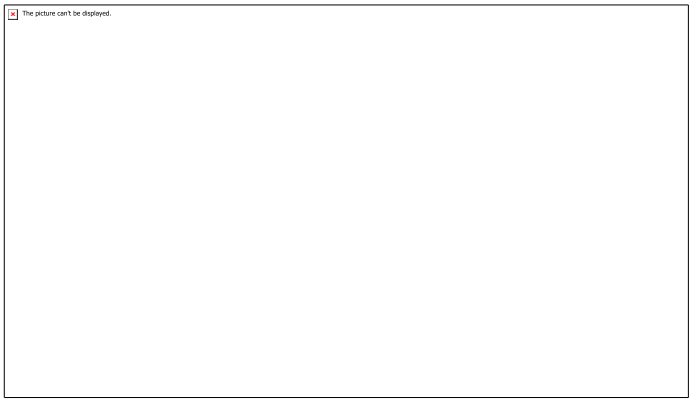
- The POS and billing systems are automated by AI to simplify the transaction process as they enable employees to pick menu items on an electronic system that automatically adds totals, taxes, and discounts.
- These systems document every sale in real time creating the correct receipts and save the data to be analyzed in future.
- It generates detailed reports about daily revenue, performance of each item and the general sales trends without any human interference.
- Smart POS solutions are connected to inventory management software to monitor the inventory levels in real time whenever a product is sold, so as to keep the records up to date.

- They minimize the possibility of human mistakes in billing, accounting, and tracking of inventory by automating calculations and data entry.
- These systems increase the speed of service especially during peak hours, by simplifying order processing to decrease the waiting time of customers.
- Incorporating transaction information, AI POS systems enable making strategic decisions concerning menu planning, pricing, stock management, and business activities in general.

3.1.5 Types of AI tools used in SME food stalls

The kind of AI tools applied in SME food stalls.

1. Petpooja



(Source: https://www.google.com/search?q=gofrugal+software&tbnid)

- Petpooja is a well known billing and restaurant management software that is applied by numerous SME food stalls.
- It is primarily utilized in billing, order control, menu control and inventory.
- This software is used to assist the owner of the food stall in knowing and comprehending the sales made daily, the commonly sold foods and their stock location bu automatic reporting and easy analysis.
- AI is used to determine the patterns of sales and performance trends.
- It is usually offers on subscription monthly and is deemed to be affordable by a majority of SME food stalls.

2. GoFrugal

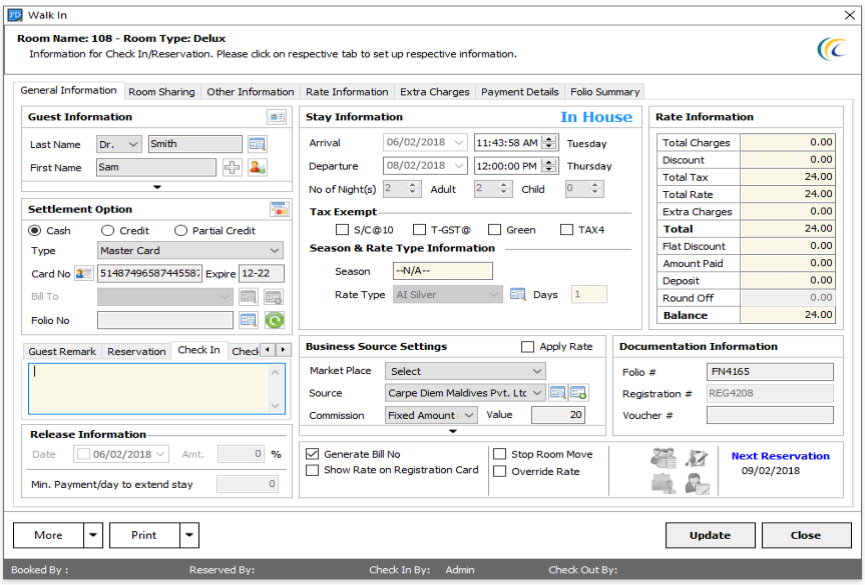
(Source: https://www.google.com/search?q=gofrugal+software&tbnid)

- GoFrugal is applied in billing, stock management, customer database and business reporting.
- The primary purpose of such software in food stalls is to help with the improved control of purchases, sales and inventory.
- Its smart capabilities enable the owners to know what is selling and the progress of the business in the long run.
- The software is provided on subscription packages and is fairly cheap to SME business.

3. Posist

- Posist is applied to order processing, billing, displaying the orders and tracking outlets performance.
- This system is used to enhance order handling and minimize errors and serve more quickly.
- AI driven reports assist owners in tracking sales dynamics and performance.
- Its cost is a bit more expensive than basic billing apps, however, remains affordable primarily to medium size stalls.

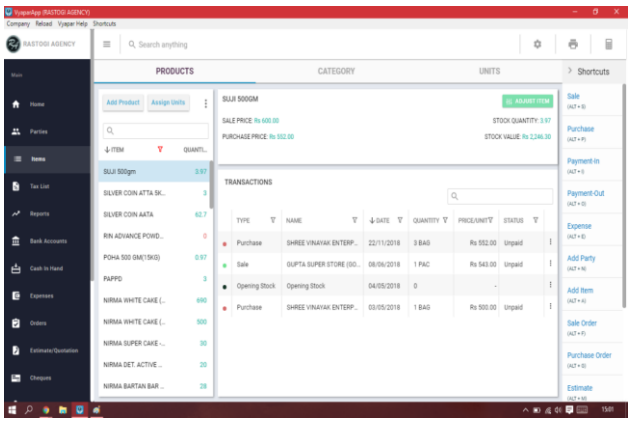
4. eZee



(Source: https://www.google.com/search?q=ezee+payment&tbnid=Y09L_d6h488U3)

- eZee offers online services to food and hospitality companies.
- It is employed in the billing and service management and business monitoring.
- This tool is to assist owners to monitor the daily operations and come up with performance reports based on smart features.
- This AI helps in demand and activity tracking.
- Its pricing is subscriptive and it is typically affordable to small food and hospitality outlets.

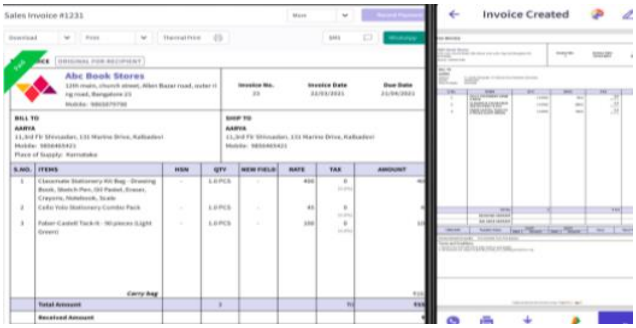
5. Vyapar



(Source: https://www.google.com/search?q=vyapar+software+images)

- Vyapar is initially applied in billing, invoicing, expense management and inventory tracking in small food stalls.
- It serves to ensure that owners manage simple and correct business records and know the daily sales and costs using automated summaries.
- The features supported by AI are sued to create fast business insights.
- The app is expensive and it is very affordable to SME owners.

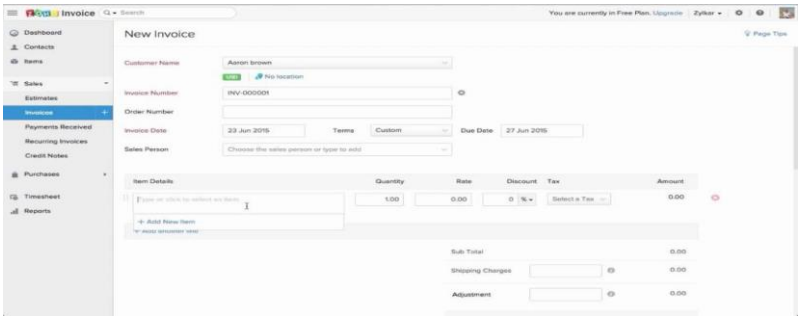
6. MyBillBook



(Source: https://www.google.com/search?q=mybillbook&tbnid)

- MyBillBook is a mobile based billing software that is applied in extremely on small restaurants..
- The purpose of this application is to simplify the process of billing every day and enable the owners to observe the sales patterns using the automated reports.
- AI operates in the background to compile the information and give easy insights.
- It is also free and low cost, and is extremely economical to SME food stalls.

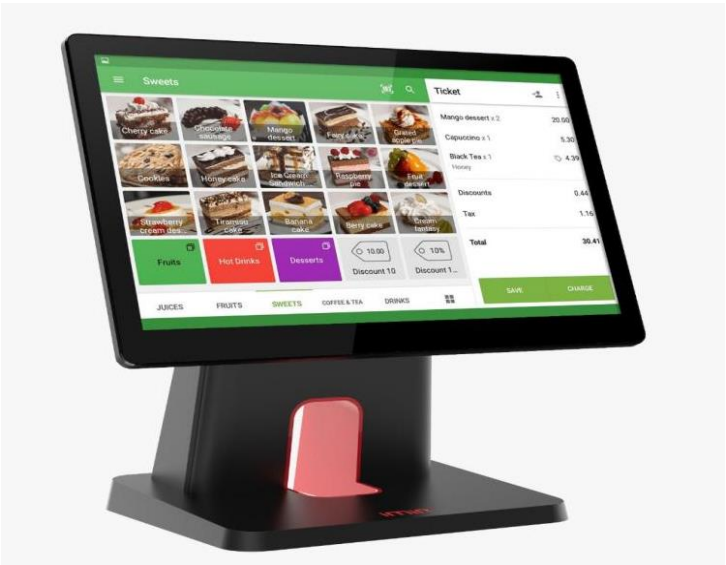
5. Zoho Invoice



(Source: https://www.google.com/search?q=zoho+invoice&tbnid)

- Zoho Invoice is primarily applied to online billing, financial administration of payments and customer databases.
- This tool is meant to assist in organized billing and adequate financial records of SME business.
- Automation using AI is beneficial in invoice creation, reminders and record keeping.
- It offers free and cheap packages and is not expensive to SME food stall owners who require an easy and professional billing services.

6. Loyverse



(Source: https://www.google.com/search?tbnid=Y09L_d6h488U3M&tbnh=0&tbnw)

- Loyverse POS is a computer based POS program designed by SME food stalls, restaurants and especially for beverages.
- It enables business owners to do billing, sales tracking and inventory management as well as employee performance using a smartphone or a tablet.
- It is also seen that the application comes with the features such as customer loyalty programs, sales reports and various methods of payments and assist in managing customers and business better.
- Loyverse can help SME food stall owners to minimize manual labour, keep proper records and make superior business decisions using the sales statistics.

3.1.6 Merits of Artificial Intelligence in SME food stalls

1. Quick billing and processing of orders

- AI helps in order processing such as fast billing through usage of AI tools and technology.
- Especially when adding GST tax automatically which helps shop keeper to increase more sales.
- This saves time to customers waiting and enhances efficiency in the services.
- Speedy billing is also helpful in assisting businesses to attend to more clients during peak demand.

2. Improves inventory and stock control

- AI helps the managers to identify the stock management whether which items are sold more and which items are sold less and also give notifications or reminders to update to restock.
- The system will automatically make changes to the stock level every time a product is sold.
- They help entrepreneurs by avoiding shortages and overstocking.

3. **Reduction in human errors**

- Mistakes in calculations or entering data may be observed occasionally in case of manual billing and record keeping.
- The AI eliminates these errors by doing everything automatically.

- The AI system will guarantee proper billing and maintenance of records.
- This enhance trustworthiness of business.

4. **Improves better day to day business**

- The business data is analyzed using AI includes the sales data and purchases by customers.
- According to this information, the system can offer valuable information to decision making.
- The owners of the business are able to determine the most popular products and the hours when sales are maximum.
- This assists them in scheduling their operation.

5. Time saving for owners

- AI does a lot of repetitive work including billing, reports, data recordings ,etc.
- This is a time saving factor to business proprietors.
- They will not have to work manually rather spend time on customer service and business growth.

6. **Understanding customer preferences better**

- AI examines the purchase behaviour of customers and points out the products that are favored.
- This gives the business owners an idea of what customers are fond of most.
- With the knowledge of the customer preferences, the businesses will be able to enhance their menu and service.
- It also contributes to the enhanced customer satisfaction.

7. **Support for digital payments and online orders**

- AI based systems embrace various types of payments like electronic wallets, cards and internet payments
- This simplifies the process of payment to the customers
- It also enhance the general business efficiency.

8. **Easy business monitoring**

- AI system offer real time sales and performance of the businesses
- The digital dashboard will allow owners to track daily purchases and income.
- This assists them to make them aware of their business operations at all times.

9. Proper sales report and data analysis

- Intelligent systems automatically create weekly, monthly or daily sales report.
- Such reports allow the owners of business to get insights into their financial performance.
- With proper reports, it is easy to monitor the profits and expenses.

10. Helps in demand forecasting

- The AI systems will be used to forecast demand by analyzing the historic sales data.
- This assists businesses in making sufficient stocks in order to meet the demands of the busy day.
- Effective forecasting prevents the possibility of stock shortage or wastage.

11. Enhance operational effectiveness

- AI can be used to automate different business processes including billing and stock management.
- This enhance the overall efficiency of doing business.
- Work is done more efficiently and in better manner.

12. Better customer service

- The use of AI assist the businesses in responding to the needs of customers in a timely manner.
- The turnaround time on billing and ordering is improved to enhance the customer experience.
- The satisfied customers will be able to come back.

3.1.7 Demerits of Artificial Intelligence in SME food stalls

1. Expensive Software and Devices.

- Smart POS, inventory software, forecasting tools, and other AI systems are costly to initially install, and many small stall proprietors cannot afford it.
- The cost of buying compatible hardware like computers, tablets, or sensors will also increase the total cost and thus adoption will be an even costlier affair.
- There are lots of small businesses that work with very thin profit margins, and it might not appear possible to spend on AI tools in comparison with old manual operations.
- Continued expenditures of software subscriptions, updates and maintenance also add to the financial burden.
- The expensive nature may deter owners and operators of SMEs to venture or implement AI or even gradually, although the long term benefits are apparent

2. Insufficient Technology Knowledge.

- Not all owners of SMEs have been exposed to the digital systems before, and thus it is not easy to learn how AI works.
- The use of technical language and complicated interfaces in AI systems may be baffling to less educated users or those who are not conversant with contemporary software.
- Lack of fundamental computer or smartphone abilities might result in errors, incorrect use, or total wastage of AI tools.
- Without adequate knowledge, stall holders can develop impediments to rely on the outside specialists, which is a costly and complicated addition to the situation.
- Learning curve of using AI systems may be a disheartening factor to small scale owners who are already doing all other business processes manually.

3. Instead of depending on the Internet and Electricity.

- Several AI applications demand a stable internet connection to operate in the clouds, update and synchronize real-time data, and this is not always present.
- The constant power outages or lack of reliable power supply in some regions may interfere with the functioning of AI and halt the daily business operations.

- Small stalls tend to stick to manual practices because it is not highly reliant on technology and utilities, which are other expenses.
- The cost of having constant power and internet connectivity may exceed the perceived advantages of small operators.
- The reliance on the technological infrastructure makes AI systems susceptible to external failure influence the overall business performance.

4. Fear of Information Loss and Data Privacy.

- The fear of hacking and misuse of sensitive sales, customer, and inventory information is the big concern raised by many SME owners about data being stored in AI systems.
- The fear of deleting digital records unintentionally or malfunctioning of the system raises the concern of the dependence on AI.
- Proprietors might be worried that such information about the proprietary business will be misused by the AI providers or third-party software.
- The absence of knowledge regarding cyber security practices enhances the lack of trust and reluctance to embrace AI tools.
- Even small violations or mistakes of AI systems can be viewed as disastrous to small businesses, which negates its implementation altogether.

5. Language Barrier

- The interface of the AI is usually built on international languages like English that might not be comprehensible to the SME owners.
- Users with limited knowledge in these languages might find it difficult to use software appropriately and decode information that AI tools are able to deliver.
- Failure to understand instructions or alternatives may cause errors in billing, updating of inventory or interacting with customers.
- The confidence of using AI can be lowered by language barriers which make the owners switch to conventional manual practices.
- he absence of localized AI tools denies the opportunity to further implement it among stall owners who would like to use software written in their native language.

6. Lack of Training

- Some AI systems are complicated and have various functions such as forecasting, managing inventory, and customer analytics, and they should be guided appropriately.
- There are very few structured training programs and tutorials on how to operate a small food stall, which are not available to many SME owners.
- The benefits of AI can be curtailed without proper training to ensure that the users utilize all the features.
- Misuse through lack of knowledge may lead to mistakes in the records of sales, stock keeping or customer information.
- Constant education and training are usually required yet the majority of small stalls are not able to pay dedicated technical support.

7. Technical Problems

- AI systems may experience glitches, bugs, or malfunctions in hardware, and that can interfere with the day-to-day running of a business.
- These technical issues may cause some valuable sales or inventory information to be lost, and this impacts decision-making.
- The owners of SMEs might lack the expertise on how to troubleshoot these problems in time, leading to frustration and losses in time.
- Technical failures often lead to the decline of trust in the efficiency of AI tools and deter regular use.
- As opposed to human operators, AI cannot know what to do instinctively or what is required in a business during system malfunctions, which poses operational risks.

8. Not Small or Temporary Stall Suitable.

- Roadside or temporary stalls tend to generate a small amount of money every day and this is not worth the expenses of installing AI systems.
- The size of operations is little, and the complex AI solutions are not needed to fulfill the routine work, such as billing or simple inventory tracking.
- Temporary stalls are usually short term or seasonal and investment in technology would not be a good pay off.
- Small stall owners might want cheaper manual solutions which are easier and faster to operate without having to learn AI tools.

- To reduce AI cost, complexity, and maintenance, it is not feasible in micro or mobile food vendors, which could restrict their adoption.

3.1.8 Outlook of artificial intelligence in small and medium enterprises (SME) in the future.

- **Introduction**
  - AI is fast changing the manner in which businesses are conducted globally.
  - SMEs are beginning to see the opportunities that AI can offer to enhance efficiency, productivity, and customer satisfaction.
  - Despite the fact that big organizations were the first to be introduced to AI technologies, SMEs are slowly embracing AI tools in conducting business because of availability of affordable digital tools.
  - It is believed that in the future, AI will become a highly important aspect of the growth and development of SMEs, making it possible to automate the processes, better the decision-making process, and enrich the customer experiences.
  - The future of AI in the SMEs is enormous because as technologies are improving, AI is becoming affordable, more accessible and more friendly to the small business owners.

1. Better AI Technologies Adoption.

- The use of AI technologies in SMEs is likely to grow in the next few years.
- With digital transformation being a key to business survival, AI-based tools and software will be adopted by many small businesses that will simply stay afloat.
- The accessibility of cloud-based AI services and low-pricing subscription plans will enable SMEs to adopt innovative technologies without having to make huge investments.
- Small businesses will use AI-based customer relationship management, sales forecasting, inventory management, and billing applications.
- Moreover, governments and other NGOs are facilitating digital takes up among SMEs through training and financial assistance.
- Such efforts will help to persuade the owners of SMEs to apply AI technologies in their everyday practice.

- With the growing consciousness over the usefulness of AI, additional SMEs will start to embrace the use of intelligent systems that will assist in streamlining their operations and overall performance of businesses.

2. Robotization of Benefit of Business.

- Automation of routine and repetitive functions will be one of the most notable changes of AI in SMEs in the future.
- Small businesses have a lot of daily tasks that involve manual work, which may take a lot of time and other resources.
- Automated tasks that will be performed by AI technologies will include billing, order processing, inventory tracking, and data entry.
- Automation will ensure that owners of SMEs concentrate on other significant areas of their work, including strategic planning and customer interaction.
- Robots with AI are more effective and efficient in repetitive tasks and can be performed without many mistakes, unlike manual tasks.
- With the increasing automation tools and their accessibility, SMEs will grow to use AI to make operations easier and more productive.

3. Data analytics result in better decision making.

- To a large extent, AI will enhance the decision-making capacity of SMEs. Contemporary companies produce substantial quantities of information regarding sales, client preferences and performance in their operations.
- Nonetheless, a larger number of SMEs do not have either the resources or knowledge to interpret such data.
- AI data analytics tools will allow the owners of SME to extract meaningful information out of their business data.
- Such systems are able to examine previous patterns of sales, the products that are popular as well as forecasting the demand in the future.
- With the predictive analytics, SMEs will be able to make more decisions concerning pricing, inventory and marketing strategies.
- The AI-driven decision support systems will enable SMEs to minimize risks and enhance profitability by offering accurate and timely information on business planning in the future.

4. Improvement of Customer Experience.

- The success of SMEs is highly dependent on customer satisfaction particularly in the competitive sector like food services and retail.

- AI technologies will greatly improve the experiences of customers, offering individual services and speedy responses.
- Virtual assistants and chatbots will be used to assist SMEs in communicating with customers via online communication and mobile applications powered by AI.
- The tools have the ability to respond to customer enquiries, give product suggestions, and help with order booking.
- Also, the AI systems are able to scrutinize the purchasing habits and preferences of the customers in order to provide personalized promotions and discounts.
- AI will enable SMEs to have better relationships with their customers and customer loyalty because it will enhance customer interaction and improve customer service quality.

5. Intelligent Inventory and Supply chain management.

- Inventory management is not always easy to SMEs and it is even more difficult in case the business is associated with perishable products like food.
- AI will be applied in the future to enhance inventory and supply chain management of SMEs by providing smart forecasting and automated tracking solutions.
- On the one hand, AI-based inventory management can track inventory changes in real time and notify the owners of a business about the shortage of supplies.
- The systems are also able to examine the sales trends to forecast the demand of certain products.
- This is because SMEs are able to keep their inventories at the optimal level as well as minimize wastage.
- The better coordination between the suppliers and the business owners will also be achieved through the AI as it will be able to give precise demand forecasts and automatically schedule orders.
- It will result in the increased use of resources and the efficiency of operations.

6. Expansion of AI-based Marketing Plans.

- Marketing is critical in the attraction of customers and sales of SMEs.

- The use of AI technologies will transform marketing strategies because it will facilitate data-driven and personalized marketing campaigns.
- The AI systems can use the demographics, buying behavior, and online activity of customers to establish target audiences.
- This information will enable SMEs to develop strong marketing campaigns that will target the right customers at the right time.
- The use of social media and digital marketing tools that are conducted using AI will enable SMEs to advertise their products more effectively and at reduced costs.
- Marketing processes like email, promotional messages and social media advertisements will also be automated by AI in the future to assist SMEs.
- These smart marketing systems will assist SMEs to enhance brand awareness and customer interactions.

7. Integration with the Digital Payment and E-Commerce Platforms.

- The growing usage of online ordering websites and electronic payments has opened new opportunities to SMEs.
- These platforms shall be further improved by AI which will offer smart transaction management systems and fraud detection systems.
- The AI-based payment system can examine the pattern of transactions and stop fraudulent activities.
- Also, AI will allow SMEs to handle online orders in a more effective manner by optimizing the delivery time and customer preferences.
- The combination of AI with e-commerce will enable the SMEs to increase their customer reach and outside their geographical locations.
- This digital divide will create new opportunities to small businesses.

3.2 Introduction of Small and Medium Enterprise (SME)

3.2.1 Introduction

SMEs are important in the economic growth of most countries particularly the developing countries such as India. Such businesses enhance economic growth and development in industries, creation of jobs, and entrepreneurship in the economy. SMEs are believed to be the pillars of the economy as they generate a lot of

employment opportunities and boost the development of the local industry. They are involved in different industries like manufacturing, services, retail, food industries and technology. They are useful in enhancing the economical framework through productivity, innovation, and improvement of the locality.

Micro, small and Medium Enterprises (MSME) are formally organized in India. This industry is controlled by the Ministry of Micro, Small and Medium Enterprises as stipulated in the Micro, Small and Medium Enterprises Development Act, 2006. This classification was introduced by the government to facilitate and encourage small businesses by offering financial aid and training and policy support. The classification of SMEs depends on the investment in the plant and machinery or equipment and the turnover per year.

SMEs are also necessary in boosting economic development since they enhance balanced growth of the regions. The large industries typically are in the urban or industrial regions and the SMEs have an opportunity to be in the rural and semi urban regions. This assists in the generation of jobs in other parts and minimizes the rural to urban migration. SMEs also make use of the local resources and manpower hence developing the local economy.

he other feature of SMEs, which is also significant, is its role in promoting entrepreneurship. A lot of people enter business ventures by small business ventures since they do not need huge capital investment. The SMEs enable the entrepreneur to test the business concepts, acquire management experience and develop over time. Many of the successful companies in the world have over time blossomed out of small businesses into giant organizations.

Besides this, SMEs introduce innovation and creativity in business environment. They are able to easily adjust to a change in market conditions and customer tastes, due to their smaller size and malleable structure. Most SMEs are launching new products and services to get new customers and also stay marketable. This enables them to react positively to economic fluctuations and technology.

SMEs also facilitate the expansion of the large industries as suppliers and service providers. Small enterprises supply many large companies with raw materials,

components, and the services of a special nature. The industrial supply chain is reinforced by this relationship, and it enhances production efficiency.

All in all, SMEs play an essential role in the economic development. Not only do they generate job opportunities but they also encourage innovation and entrepreneurship and development in the region. They perform a significant role in the national economy due to their contribution to the industrial production.

3.2.2 Concept of Small and Medium Enterprises

Small and Medium Enterprises is a notion that is used to refer to enterprises operating relatively at smaller levels than the large corporations. These are firms that have minimal investment, small number of workers, and average production capacity. The concept of SMEs is aimed at the development of small-scale industrialization and economic entrepreneurship of the population who might not afford huge capital outlay.

SMEs are thought to be a significant constituent of the economic system in that they aid in the effective utilization of local resources. They work in various industries including manufacturing, services and retail trade and technology. The concept of SMEs points to the significance of promoting small business such that it can help the economy to grow and reach the goal of creating employment.

The related concept of SMEs in India is the development of the Micro, Small and Medium Entrepreneurships (MSME) segment. This grouping assists the government to formulate suitable policies and assistance programs to various types of businesses. Micro enterprises are the smallest forms of businesses that have minimal investment and turnover. Small businesses have slightly higher investment and production scale whereas the medium businesses are those firms which have a bigger scale of business in the SME segment.

SMEs are also a concept that aims at enhancing entrepreneurship and innovation. Small businesses have been known to give people a chance to venture and acquire a business skill. Most entrepreneurs start with small businesses and grow slowly with the amount of experience and financial strength they have.

Flexibility and adaptability is another concept that is significant to the SME concept. In contrast to big corporations, SMEs possess less complicated organizational systems and levels of management. This enables them to take fast decisions and react fast to the market demand. Their flexibility will allow them to launch new products, enhance services, and embrace new business practices.

This is also the contribution that SMEs have to the large industries and enhancement of supply chains. Numerous small businesses become the suppliers of raw materials, components and servicing in large businesses. This collaboration between the small and large businesses assists in enhancing the productivity and efficiency in industries.

Moreover, the SMEs concept also emphasizes the role of SMEs in ensuring inclusive economic growth. SMEs facilitate the standard of living of many people in the society by providing them with job opportunities and stimulating entrepreneurial activity in the local areas. They also help in the growth of the rural and semi urban environment because they set up businesses in areas where giant industries do not exist.

Comprehensively, SMEs are developed on the concept of small business ventures that help in boosting economies, creation of employment opportunities and innovation. Policies, financial aid and training programs can be made to support SMEs to boost economy and enhance sustainable development.

3.3.3 Role of Small and Medium Enterprises

1. Employment Generation

- SMEs generate numerous working positions in various industries.
- They hire skilled and unskilled workforce.
- They assist in avoiding unemployment within the economy.
- They offer employment both in rural and urban regions.
- They are sustaining the lives of several families.

2. **Entrepreneurship Promotion.**

- SMEs motivate individuals to venture into their businesses.
- They contribute to the acquisition of entrepreneurial skills.
- They encourage self-employment.
- They encourage people to be innovative and risk takers.
- They assist young entrepreneurs to venture into the business.

3. **Economic Development**

- SMEs add on to the national income.
- They favour industrial production.
- They boost the economic activities in the country.
- They assist in the general economic development.
- They boost the economy of the country.

4. **Evenly distributed Development.**

- SMEs facilitate rural development.
- They provide employment in the small towns and villages.
- They decrease the urbanization.
- They contribute to economic growth in the regional areas.
- They enhance economic balance of the region.

5. **Support to Large Industries**

- SMEs provide raw materials to the bigger industries.
- They supply the parts and components of production.
- They provide services of support to big companies.
- They enhance industrial collaboration.
- They assist in the smooth production processes.

6. **Innovation Promotion.**

- SMEs come in with fresh ideas and products.

- Creative business solutions are encouraged by them.
- They test new forms of business.
- They enhance products as per the needs of the customers.
- They facilitate market creativity.

7. Impact of Local Resources.

- SMEs make use of raw materials available locally.
- They employe local workers.
- They use available local talents and skills.
- They encourage the utilization of local resources.
- They favour the local economic activities.

8. Export Promotion

- SMEs manufacture products to the global markets.
- They have the effect of increasing foreign exchange income.
- They assist in popularization of Indian products in the world.
- They increase the trade opportunities.
- They promote expansion in export trade.

9. Development of Skills

- SMEs offer real life training to employees.
- They assist employees in developing new skills.
- They enhance technical learning.
- They promote experience in learning.
- They produce competent human resource.
- Increase in the Standard of Living.
- SMEs enhance the income opportunities.
- They assist families to have a better lifestyle.
- They alleviate poverty in most areas.
- They promote favourable living standards.
- They enhance the general quality of life.

10. Industrial Growth

- SMEs enhance the production of goods and services.
- They help in the growth of industries.
- They boost the manufacturing industry.
- They contribute to the growth of the industrial activities.
- They play their role in industrial development.

11. Improving Supply Chains.

Larger companies obtain materials supplied by SMEs.

They promote logistics and delivery.

- They establish a continuous flow of production.
- They enhance the coordination of businesses.
- They assist industries to be efficient.
- Encouraging Competition
- SMEs enhance competition in the market.
- They motivate companies to be of better quality.
- They help reduce monopolies.
- They ensure equitable product pricing.
- They encourage businesses to be creative.

12. Digital Transformation Support.

- SMEs embrace the use of digital payment systems.
- Online marketing and e-commerce are used by them.
- They adopt electronic management systems.
- They embrace new technologies such as AI.
- They enhance productivity by use of digital systems.

3.3.4 Evolution of Small and Medium Enterprises

- **Pre 1947 - Traditional Small Industries.**

The small businesses in India before independence were mostly traditional ranging through handcraft, handloom weaving, pottery and small food production. These

industries were predominantly located in the rural setting and they provided jobs to the locals.

- **1948 - First Industrial Policy**

Small-scale industries became an important part of economy development and employment generation in the Government of India through the introduction of the Industrial Policy of 1948.

- **1951 - Industries (Development and Regulation) Act.**

The Industries (Development and Regulation) Act, 1951 came into being to control industrial growth in India, and the small-scale industries.

- **1955 - Support Organizations were created.**

The government came up with organizations like the National Small Industries Corporation to offer financial aid, marketing services and raw materials to the small business owners.

- **The industrial policy resolution of 1956.**

Industrial Policy Resolution of 1956 accentuated the development of small industries and encouraged the regional balancing development.

- **The introduction of Village Industries was promoted in 1957.**

Khadi and Village Industries Commission was set up in order to develop village industries and offer assistance to rural entrepreneurs.

- **1977 - Priorities towards Small-Scale Industries.**

The Industrial Policy of 1977 accorded particular priority to small-scales industries and provided policies that promoted rural and cottage industries.

- **1991 - Economic Liberalization**

In 1991, India had implemented significant economic reforms. These reforms added competition to the market and made SMEs strive to enhance efficiency, productivity and quality of products.

- **The growth of Technology and IT SMEs, 2000.**

The early years of 2000s witnessed a significant number of SMEs beginning to employ the use of information technology, computers and digital systems to enhance their business processes.

- **2006 - MSME Development Act**

One of the greatest achievements was the implementation of the Micro, Small and Medium Enterprises Development Act, 2006. According to this act, the enterprises were officially divided into the groups of micro, small, and medium and were better supported by the government.

- **The promotion of the manufacturing sector took place in 2014.**

To boost production and facilitate the development of SMEs, the government introduced the project, Make in India.

- **2015 - Digital India Initiative**

Digital India was introduced by the government to provide the promotion of digital technologies and the use of online platforms and digital services by small businesses.

- **2020 - Support During COVID-19**

The government initiated the Atmanirbhar Bharat Abhiyan during the pandemic to fund and give loans to SMEs and revise the policies to support the businesses.

- **2020 - New MSME Classification**

In order to promote the development of small and medium enterprises, the government restructured the MSME classification according to the investment and turnover.

- **Current - Digital and Artificial Intelligence adoption.**

The digitization of SMEs through the use of digital technologies today is evident in the adoption of e-commerce platforms, online marketing, digital payments and artificial intelligence by many SME to enhance efficiency and increase their market presence.

3.3.5 Advantages of SME

1. Employment Generation

- **Generates numerous employment opportunities -** SMEs need employees to work in various areas like production, sale and services and this aids in generating numerous job opportunities.
- **Lowers unemployment -** SMEs create employment opportunities to a large number of people and hence lowering the rate of unemployment in the economy.
- **Hires both skilled and unskilled laborers -** SMEs give chances to high-skilled professionals and individuals who lack specialized knowledge.
- **Generates revenue to families-** Jobs generated by SMEs enable families to get a steady income and live better lives.
- **Supports local jobs -** A large number of the SMEs employ the local people and this fortifies the local economy.

2. Low Capital Requirement

- **Need not attract too much investment -** SMEs will be able to commence operations with a relatively small amount of financial resources as compared to large industries.
- **Easy to start-** Due to the amount of investment to be made, people can easily start small businesses.
- **Fitted in small businesses -** Individuals who have limited funds can start SMEs and grow them step by step.
- **Low financial risk -** With the small investment, financial risk is also minimized compared with the big businesses.

- **Does not need huge initial capital -** SMEs are capable of initiating operations by using minimal equipment and tiny infrastructure.

3. Promotion of Entrepreneurship.

- **Encourages individuals to become entrepreneurs -** SMEs make people start businesses and become entrepreneurs.
- **Grows business -** An SME makes people get to know how to manage, plan finances and do marketing.
- **Fosters innovation -** Small businesses tend to bring new products and innovative ideas in a business.
- **Develops leadership skills -**Running a business will enable people to acquire leadership and decision-making skills.
- **Encourages self employment -**SMEs enable people to generate their own employment rather than relying on employment.

4. Flexibility in Operations

- **Adapts to changes in the market easily -** SMEs are able to change their strategies fast when the market conditions are altered.
- **Fastness to customer demands -** Small businesses find it easy to alter their services depending on customer response.
- **Is able to change products easily-** SMEs are able to add new products or alter the existing products without complex processes.
- **Simple business structure -** SMEs are less complex in terms of the number of management levels and therefore they are easy to run.
- **Simple operations -** The owners can readily oversee the business operations due to their smaller size.

5. Quick Decision Making

- **Quick decision making -** Small management teams or owners can make decisions fast.
- **Red tape -** SMEs do not need a lengthy decision-making process.

- **Immediate problem solving-** Problems are recognized and solved on short notice.
- **Fast business decision-making -** The SMEs are able to make instant decisions when new opportunity comes.
- **Improved operational efficiency -** Swift decisions enhance the rate of productivity and business output.

6. Balanced Regional Development.

- **Boosts rural development -** SMEs develop rural and semi urban areas by setting up businesses in them.
- **Generates employment in the small town -** A large number of SMEs are located outside the big cities and generate employment there.
- **Reduces migration to the city -** Local jobs eliminate the necessity of people relocating to urban locations.
- **Helps in local businesses -** SMEs promote the development of other businesses within the area.
- **Boosts local economy -** There is enhanced economic activity in the region in terms of employment and trade.

7. Innovation and Creativity

- **Promotes novelty -** SMEs are known to be experimental towards novel ideas.
- **Market innovative products -** Small businesses will bring innovative and tailored products.
- **Easily experimental -** SMEs can easily test new products or services.
- **Launches distinctive services -** A lot of SMEs innovate distinctive services in order to appeal to the customers.
- **Favours innovation -** Constant innovation makes SMEs stay competitive in the market.

8. Support to Large Industries

- **Supplies raw materials -** The SMEs supply the raw materials or parts needed by the big industries.

- **Provides support services -** They are providing services i.e. logistics, packaging or maintenance.
- **Empowers supply chain -** SMEs aid to sustain a continuous flow of materials in the process of production.
- **Assists in the expansion of the large industries -** SMEs aid in the expansion of the larger firms through supporting the production activities.
- **Develops business co-operation -** Cooperation between the small and large business enhances the industrial sector.

9. It has Contributed to Economic Growth.

- **Increases national income -** The SMEs manufacture goods and services that are part of the GDP of the country.
- **Provides jobs -** SMEs create jobs and this helps to boost the economy.
- **Favors industrial development -** SMEs are major actors in boosting the industrial industry.
- **Enhances exports -** A large number of SMEs engage in production of goods they export to foreign countries.
- **Economical empowerment -** With the growth of SMEs, the general economy is boosted.

3.3.6 Disadvantages of SME

1. Limited financial resources

- Here it is difficult to get loans because banks hold back or delay to give loans to some SME like micro enterprises and small enterprises due to limited collateral if the loans are not paid.
- Most of the SME has small and micro enterprises which results in limited capital.
- If there is any economic recession or inflation, the SME gets only limited funds. This leads to affect the country GDP.
- Due to low funds it is difficult to expand their business.
- Most of the SME rely on personal savings which acts as a safety.

2. Poor Technology

- Due to limited funds, SME has poor technology where the adoption and buying of tech tools are at very high cost.
- Some SME have oldest or outdated technology which is not reliable to the business and by using it cause low performances.
- More investment is needed to purchase high and modern technology which is not access to most of the SME where it has micro and small businesses.
- The usage is considered to be very low because of the low and few funds by SME.
- Usage of less technology affects both efficiency and performances of SME.

3. Limited mangerial skills

- Most of the SME owners may not have knowledge of managerial skills due to lack of training.
- Lack of planning and budgeting leads to loss in business where SME owners has less exposure due to improper training.
- Sometimes less marketing knowledge leads to low productivity where customers feel dissatisfied with their services.
- If there is no proper strategies and experiences which cause more errors in the business.
- The most important challenge in managerial skills is to deal with the employees which is the greatest managerial task.

4. Heavy Competition

- Most of the SME contains big companies and the competition is very high because other big companies have good will and recognized across the world and also have adequate resources.
- Most of the SME have limited funds which the major drawback for brand recognition through advertisements, where promotion and advertisement have more cost and its is very difficult for SME to adopt.
- And also customers prefer high quality, good will, safety and popular brands which affect some SME's.

- One of the major barrier is high competition reduces shares in the market.

5. Limited globalization

- SME is not a globalized business where it often deals with selling products and giving services to local customers.
- SME does not have large distribution networks like KFC, Domino's and Samsung. They have local services to the customers.
- If SME wants to reach globally they need additional funds and resources but it is difficult for SME to have more funds.
- Most of the sales are from local customers because SME are dealing with local services market.
- Lack of promotion and advertisement is a major drawback here.

6. Supply Chain Problem

- Due to lack of investment in the supply of raw material, SME face delays in getting the materials on time.
- Sometimes the transportation problems like unpredictable weather conditions may result in slow movement of gods.
- There is very few number of suppliers due to less funds which is also result in slow movement of goods.
- Transportation cost is very high for SME.
- Inventory problem is the major drawback in supply chain management, where goods or products get damaged due to certain conditions.

7. Lack of skilled employees

- If there is lack of skilled employees in SME this results in lack of efficiency and productivity and the outcome is very low.
- In SME, there is no professional training programs conducted as it affects the development of the business.
- There is difficult in hiring skilled workers in SME.
- Many skilled workers hesitate to work in SME because of low salary problems and less financial benefits.

- High employee turnover is the major drawback here, if employees are not satisfied they may leave the business.

8. Lack of R&D

- Due to lack of funds there is no proper research and development in SME which results in lack of research.
- In SME, lack of research leads to low quality of product and services which does not satisfy the customer needs.
- Limitation of research leads to less or poor innovation.
- SME has low funds and resources where it affects the growth of technology and development.
- Every product and services should be tested, but in SME the experiments are avoided which is the major drawback.

9. Difficulty in adopting new technology

- In SME, lack of technology knowledge and AI adoption is the major limitation, because of less training to improve skills.
- The next drawback is the high cost of installation of digital and AI tools, SME could not afford expensive tools.
- Some SME owners were not aware of AI or digital tools, their usage and role of the tools in their business.
- Most of SME chooses traditional methods than modern updated method.
- Training is the best effective solution for upcoming these barriers.

3.3.7 Types of SME

Small and Medium Enterprises (SME) plays a vital role in our country. They are classified under Medium, Small and Medium Enterprises(MSME) sector and it is regulated by Ministry of MSME Act 2006.

SME is classified under different types based on the enterprise size, nature of activity and ownership structure.

- **Classification of SME based on Size**

SME is classified into Micro enterprise, Small enterprise and Medium enterprise under based on size

1. **Micro Enterprise**

- Micro Enterprise is a very small enterprise among the classification of SME.
- They have a very small investment and level of manpower.
- According to MSME, under the Micro enterprise, the investment in plant and machinery or equipment should not more than Rs. 2.5 crore and Annual turnover not more than Rs. 10 crore. (*source:)*
- Micro enterprise is divided into small business such as tailoring, home made bakery, small food stalls, food truck and road side shops.
- They are owned either by sole proprietorship or small family.
- They are operated both in rural and urban areas.

2. **Small Enterprise**

- Small Enterprise is moderately big than Micro enterprise.
- They have a higher investment and level of manpower.
- According to MSME, under the Small enterprise, the investment in plant and machinery or equipment should not more than Rs. 25 crore and Annual turnover not more than Rs. 100 crore.
- Small enterprise is divided into mall restaurants, small IT companies and printing presses.
- Small enterprise is very vital and important as it increases employee opportunities.

3. **Medium Enterprise**

- Medium Enterprise is the largest in MSME.
- They have a very higher investment and higher manpower.
- According to MSME, under the Medium enterprise, the investment in plant and machinery or equipment should not more than Rs. 125 Crore and Annual turnover not more than Rs. 500 crore.

- Medium Enterprise is divided into large scale food processing and IT sectors.

➢ **Classification of SME based on Nature of activity.**

SME is classified based on the activity of business.

1. Manufacturing enterprise

- Manufacturing Enterprise is the enterprise where it involves activities like manufacturing and production of goods and services.
- Here, they convert raw materials into finished products and sold them in market.
- The Manufacturing enterprise involves and plays an important role in industrial sectors where it has activities such as production of products and goods.

2. Service enterprise

- Service Enterprise is the enterprise where it involves activities like providing services to the customers.
- The services are like hospitality, delivery services, IT sectors, and beauty parlour.
- Nowadays SME adopts AI to work efficiently with digital transformation.

➢ **Classification of SME based in Ownership structure**

SME is classified based on the ownership of the businesses

1. Sole proprietorship

- Sole proprietorship is a type of business where it is owned and managed by the owner or single person.
- The owner is liable of its own actions whether it is a profit earned or loss attained and also if their any risk.
- There are many Sole proprietorship SME in India because it is easy to open with low cost, expenses and less legal formalities.

2. Partnership Firm

- Partnership is a type of business where it is owned and managed by two or more persons who agree on partnership terms and conditions.
- The profit, loss and risk is shared among partners with a certain ratios.
- In SME, the partnership firm is divided into small restaurants and small manufacturing firms.

3. Private limited company

- Private limited company is a type of company where it has more structured form of business.
- It is registered under company law and separate legal entity, where it is mandatory to follow the rules.
- In Partnership, the company divided into shares by shareholders.
- In SME, the Private limited company needs more adequate capital to run.

4. Limited liability Partnership

- Limited liability partnership is a type of company where it combines both private limited companies and partnership firms.
- They work under legal terms and conditions
- The liability is limited, where it involves all companies together and the private asset is protected.

3.3.8 Government support programs for SME in adoptin AI

1. Digital India Initiative

The Digital India program initiated by the Government of India is one of the most relevant programs that facilitate the use of AI among SMEs. The primary aim of this project is to make India a digitally empowered nation and knowledge economy.

Digital India advances the application of digital technologies in many industries such as the business sector, education, healthcare, and governance. Within

the framework of this initiative, SMEs will be motivated to use digital devices, including cloud computing, artificial intelligence, and online payment systems.

In case of SME food stalls, Digital India has facilitated the popularity of digital payment system including UPI, QR codes, and mobile wallets. The sales data of these digital transactions can be analyzed using AI-powered applications to generate information about customer behavior and buying habits.

Digital India also aids in the development of digital infrastructure and internet connectivity, which is the key to the implementation of AI technologies by small businesses.

2. IndiaAI Mission

Government of India has also initiated the IndiaAI Mission to enhance the creation and adoption of Artificial Intelligence in the country. This program is aimed at AI infrastructure creation, innovation, and AI technologies research and development.

IndiaAI Mission is a project to establish a robust AI ecosystem through the provision of computing resources, datasets and start up and enterprise funds. Mission is also aimed at educating people and business owners in AI-related skills to become more digital and technologically empowered.

Government programs have also been launched to educate millions of entrepreneurs and workers in the country on AI. Certainly, in such a case, the training of AI programs has been proposed via Common Service Centres to enhance digital literacy skills among rural entrepreneurs and small business owners.

Through such training, the owners of SMEs will learn to consider AI in managing businesses, serving customers, and marketing their products.

3. MSME Champions Scheme

The scheme is the MSME Champions which is a technology-based program that was introduced by the Ministry of Micro, Small and Medium Enterprises as a

way of assisting small businesses. This platform makes it a centralized system through which SMEs can be guided, solve business issues and information about government schemes can be accessed.

The portal will incorporate new digital solutions such as artificial intelligence and data analytic to help SMEs to overcome operational and financial problems. It is a one-stop centre that SME owners could find help regarding production, technology usage, marketing, and financing.

In the case of SME food stalls, this platform assists the entrepreneur in gathering information about digital tools, training options, and financial funding that would allow integrating AI-based technologies.

4. Atal Innovation Mission

The other significant program that promotes technological innovation in India is the Atal Innovation Mission. The program was introduced to encourage innovation, entrepreneurship, and start up development in the country.

The mission provides incubation centers, research facilities and start-up ecosystems which deal with the development of advanced technologies such as the Artificial Intelligence. These innovation centers will help SMEs and start-ups to get mentor ship, funding options, and technical assistance.

These innovations can be useful to the owners of food stalls and small restaurant businesses as they can use AI-based systems to manage the orders, study the customers, and manage the supply chain.

5. Samarth Udyog Bharat 4.0

The project of Samarth Udyog Bharat 4.0 aims at increasing the adoption of Industry 4.0 technologies among Indian industries (including MSMEs). The term Industry 4.0 is used to denote the application of new technologies like Artificial Intelligence, robotics, big data, and the Internet of Things in business processes.

Within the frames of this program, awareness campaigns, training programs and demonstration plants are arranged to make the SMEs know the advantages of the modern technologies. These hubs offer empirical information on the way AI and automation can enhance the efficiency of production and business productivity.

Though the program is primarily concerned with manufacturing industries, its strategies of digital transformation are applicable to the service industries like food industries as smart technologies are encouraged to be used.

6. SMEs financial support programs.

Funding is a very important factor in facilitating SMEs to embrace new innovative technologies like Artificial Intelligence. The government has also initiated a number of financial programs to avail easy access to credit and lessen financial impediments on the small businesses.

The Credit Guarantee Fund Trust on Micro and Small Enterprises (CGTMSE) has become one of the key programs. This plan offers small businesses loans without collateral so that they can invest in technological up-grading and digitalization.

Through these kinds of schemes, SME food stalls owners will be able to invest in AI-based POS systems, inventory management software, and digital marketing tools using financial aid.

Also, the small industries Development Bank of India (SIDBI) provides special loans and financial schemes to the SMEs to take up technology with the help of government-sponsored financial institutions.

7. Skill Training and Artificial Intelligence Training.

Another major area of government support in the use of AI technologies by SMEs is skill development. A number of SME owners do not have technical knowledge regarding AI tools and digital mechanisms, which makes them incapable of using such technologies.

To curb this problem, the government has initiated a number of training initiatives that would enhance digital literacy and technical expertise among the entrepreneurs. Such programs consist of workshops, online courses, and certification programs concerning artificial intelligence, data analytics and digital marketing.

The practical knowledge in the application of AI in business activities is taught in training programs organized in Common Service Centres and skill development institutes. Such programs make entrepreneurs know how AI tools can assist in sales analysis, inventory forecasting and customer relationship management.

8. Technology MSMEs Centers.

The government is also coming up with the advanced technology centers that will assist the SMEs to embrace modern technologies. These facilities offer training, support in the form of research and accessibility to superior tools and equipment.

Indicatively, some are new technology centers being established in various regions of India to enhance the technological capacities of the MSMEs. These are the centers where the small businesses can be trained on automation, robotics, and digital technologies to implement modern solutions.

9. Online Solutions to SMEs.

The role of the government in the adoption of AI by SMEs is also significant because the government has developed digital platforms previously. These websites have information, training tutorials and access to government schemes.

As an illustration, the collaboration of the technology firms with government agencies is aiding in creating AI-based digital platforms that advise SMEs in embracing digital technologies. There are also platforms which have AI bot chatbots that offer real-time access to information on government programs, financial assistance and digital resources on small businesses.

3.3.9 Key Announcements of Budgets 2026 on SMEs.

1**. 10,000 Crore SME Growth Fund**

The establishment of a [?]10,000 crore SME Growth Fund is one of the most notable ones in the Union Budget 2026. This fund will support high potential SMEs financially; they can grow and compete both nationally and internationally.

The fund will help SMEs by:

i. Getting access to growth capital.
ii. Promotion of innovation and the use of technology.
iii. Empowering business growth and scaled up.
iv. Enhancing equity investment access.
v. With this initiative, the SME ecosystem will be empowered, and it is likely to help in building globally competitive enterprises.

2. **2000 Crore Top-Up to Self-Reliant India Fund.**

The Self-Reliant India Fund (SRI Fund) has also received more funding by the government 2,000 crore.

This fund aims at promoting micro and small businesses in the form of equity and financial support. It assists the business which requires financing and may not be able to get a conventional bank loan easily.

The initiative has been found to be useful especially to:

i. Startups
ii. Small-scale manufacturing enterprises.
iii. Small restaurants and other service businesses like food stalls.

3. **Enhancing Availability of Credit and Finance.**

One of the biggest problems of SMEs is access to finance. In the Union Budget 2026, there are a number of initiatives aimed at enhancing access to credit by the small businesses.

Key initiatives include:

i. The credit guarantee schemes are to be expanded.
ii. Small business loans without collaterals.
iii. MSMEs have their loans processed faster.

These financial supports assist the entrepreneurs to invest in technology, infrastructure and business development.

4. **Upgrading the TReds Payment System.**

The Trade Receivables Discounting System has also been enhanced by the government to enhance cash flow in SMEs. This system enables the SMEs to get payment quicker on the supply of goods and services to large corporations or government organizations.

In the new budgetary proposal:

i. TReDS are required to be used by large companies and government to make payments to SMEs.
ii. It will be integrated with Government e-Marketplace (GeM) to enhance transparency and efficiency of payments.
iii. This program decreases payment delays and enhances working capital of small businesses.

5. Streamlining of Taxation and Compliance processes.

The budget 2026 also aims at cutting back on regulatory requirements on SMEs. Most owners of small businesses are faced with complex taxation processes and rules.

To overcome such problem, the government implemented such measures as:

i. Streamlined GST business processes.
ii. Increased tax bracket rates on small business.
iii. Extended deadlines on revised tax-filing.

These reforms are meant to ease the operations of SMEs and minimise the administrative expenses.

6. Digitalization and Technology Adoption.

The other central area of the budget is to make SMEs embrace the use of modern technologies like artificial intelligence, digital payment systems, and cloud-based business management tools.

The government expects to promote digital transformation by:

i. Technology support services.
ii. The development of digital infrastructure.
iii. Entrepreneur skill training.

With the use of digital tools, SMEs will be able to be more productive, lower the operations costs and improve customer services.

7. Building of Leader MSMEs.

The Union Budget 2026 will establish the so-called Champion MSMEs that will be able to compete on the international level and promote economic development. This project is aimed at supporting the growth of promising SMEs and their move to the international markets.

The program assists the businesses by:

i. Making financial support.
ii. Promoting export business.
iii. Encouraging the development of innovation and high-technology.

Analysis and Interpretation

The purpose of this chapter is to analyze the role of AI among SME food stalls in Chennai City. Primary data was collected from 200 SME food stall owners. The data was analyzed using SPSS software. The selected tools like Pearson correlation, One way Anova, Simple linear Regression and Regression.

4.1 Cronbach's Alpha Reliability Statistics

Table 4.1 Reliability Statistics

Cronbach"s Alpha	No. of . Items
0.888	50

The above table shows that the reliability score obtained using SPSS Cronbach's Alpha test is 0.888, which is greater than the standard value of 0.6. As a result, the constructed questionnaire is acceptable and reliable.

4.2 Descriptive Analysis

Table 4.2 Demographic Table

Demographic	Groups	Frequency	Percent
Gender	Male	187	93.5
	Female	13	6.5
	Total	**200**	**100**
Age (in years)	Below 25 years	8	4.0
	25-35 years	91	45.5
	36-45 years	62	31.0
	46-55 years	31	15.5
	Above 55	8	4.0
	Total	**200**	**100**
Educational Qualification	School level	162	81.0
	Diploma	33	16.5
	Undergraduate	5	2.5

	Total	**200**	**100**
Type of food stall	Road side food	24	12.0
	Small Restaurant	50	25.0
	Bakery	50	25.0
	Food Truck	25	12.5
	Beverage-focused stall	51	25.5
	Total	**200**	**100**
Nature of Ownership	Sole proprietorship	181	90.5
	Partnership	17	8.5
	Family business	2	1.0
	Total	**200**	**100**
Years of business experience	Below 3 years	25	12.5
	3-5 years	122	61.0
	6-10 years	49	24.5
	Above 10 years	4	2.0
	Total	**200**	**100**
Number of employees	Below 5 years	147	73.5
	5-10	42	21.0
	Above 10	11	5.5
	Total	**200**	**100**
Average daily sales	Below 5000	69	34.5
	5001-10000	106	53.0
	10001-20000	25	12.5
	Total	**200**	**100**
Currently using devices	Smartphone	62	31.0
	Desktop	133	66.5
	Digital menu display	5	2.5
	None of the above	0	0
	Total	**200**	**100**
Activities done by	Billing and payment	122	61.0

suing digital apps	Sales or profit tracking	6	3.0
	Digital payments	48	24.0
	I do not use any digital app	24	12.0
	Total	**200**	**100.0**
Digital apps and tools	Petpooja	35	17.5
	Go Frugal	37	18.5
	Whatsapp business	9	4.5
	Loyverse	41	20.5
	MyBillBook	5	2.5
	Swiggy	6	3.0
	Phone pay/ Gpay	43	21.5
	I do not use any app	24	12.0
	Total	**200**	**100.0**

Here, The data obtained is presented in a meaningful, easy to understand form through the percentage analysis. It assists in knowing the spread of respondents based on various sections of the population like the gender, age, educational level, nature of food stall, ownership pattern, business experience, number of employees, daily sales, devices utilized, activities carried out with the help of digital applications, and digital tools accepted. The outcome is that most of the SME food stall businesses are run by men since 93.5 percent of the respondents are men and only 6.5 percent are women.

With respect to age group, most of the respondents are in the 25-35 years age bracket (45.5%), then 36-45 years (31.0%), 46-55 years (15.5%), and below 25 years and above 55 years (4.0%), respectively.

Concerning the level of education, the majority of the respondents (81.0) have attended school level, 16.5% possess diploma, and only 2.5% possess undergraduate level education.

According to type of food stall analysis, beverage-driven stalls take 25.5 percent, small restaurants and bakery take 25.0 percent, food trucks take 12.5 percent, and roadside food stalls take 12.0 percent.

In terms of the type of the ownership, most of them (90.5) are a sole proprietorship, a partnership (8.5), and a family business (1.0).

Regarding business experience, 61.0% of the respondents have an experience of 3-5 years, 24.5% have an experience of 6-10 years, 12.5% have a below 3 years experience, and only 2.0% have the experience of 10 years or more.

The breakdown of the employees indicates that, 73.5 percent of food stalls have less than 5 employees, 21.0 percent have between 5-10 employees, and 5.5 percent have more than 10 employees implying that a majority of SME food stalls are run on small scale.

Regarding average daily sales, 53.0% of the respondents make an average of between 5,001 and 10,000 a day, 34.5% make less than 5,000 and 12.5% make between 10,001 and 20,000 a day.

On the use of devices, 66.5% of the respondents use desktops, 31.0% use smartphones and 2.5% use digital menu displays.

The tasks carried out with the digital application reveals that 61.0% make use of apps in billing and payment, 24.0% in digital payment, 3.0% in sales or profit tracking, and 12.0% do not use digital application at all.

The last category of analysis of digital apps and tools utilised shows that PhonePe/GPay (21.5) and Loyverse (20.5) are the most frequently used, as well as GoFrugal (18.5) and Petpooja (17.5) have the lowest percentages, with smaller percentages utilizing WhatsApp Business (4.5), Swiggy (3.0), and MyBillBook (2.5).

Therefore, percentage analysis is suitable to interpret the demographic picture and digital adoption tendencies of the SME food stall owners in the region of the study.

4.3 Hypothesis Testing

Hypothesis 1: Impact of Artificial Intelligence in enhancing the overall operational performance of Small and Medium Enterprises food stalls.

The following hypothesis is used to find the relationship between the usage of AI tools and operational performance of SME food stalls in Chennai.

H0: There is no significant relationship between the use of AI tools and operational performance of SME food stalls.

H01: There is a significant relationship between the use of AI tools and operational performance of SME food stalls.

Table 4.3.1 Impact of AI in overall operational performance using Regression

Anova

	Model	Sum of squares	df	Mean Square	F	Sig.
1	Regression	10.646	1	10.646	1.648	.201
	Residual	1279.354	198	6.461		
	Total	1290.000	199			
Source: *Primary Data*						

Coefficient

Model	Unstandardized Coefficients		Standardized Coefficient	t	Sig.
	B	Std. Error	Beta		
(Constant)	21.248	.266	.091	79.835	.000
AI tool usage	.071	.055		1.284	.201
Dependent Variable: *Overall Performance*					
Source: *Primary Data*					

Interpretation

The results shows from the above table there is no relationship between awareness and adoption of AI tools and the overall performance of SME food stalls in Chennai City. The Regression result (Sig=.201) shows that the awareness and adoption of AI does not significantly influence the overall performances.

Inference

Since the significant value is less than 0.05, the null hypothesis is rejected and the alternative hypothesis is accepted. Therefore the awareness and adoption of AI plays a significant role in improving the overall performance of SME food stalls in Chennai City.

Hypothesis 2: Awareness and Adoption of AI tools among SME food stall owners in Chennai City.

The following hypothesis is used to find the relationship between the awareness and adoption of AI tools among SME food stall owners.

H0: There is no significant difference between the level of awareness and adoption of AI tools among SME food stall owners.

H02: There is significant difference between the level of awareness and adoption of AI tools among SME food stall owners.

Table 4.3.2 Awareness and adoption of AI tools using One way Anova

	Sum of squares	df	Mean Square	F	Sig.
Between Groups	78.419	3	26.140	17.578	.000
Within Groups	291.456	196	1.487		
Total	369.875	199			
Dependent Variable: *Awareness and Adoption*					
Source: *Primary Data*					

Interpretation

The results shows from the above table is a positive relationship between awareness and adoption of AI tools among SME food stall owners in Chennai City. The Anova result (Sig=.000) shows that when awareness increases, the adoption of AI is also increases. Further, the Anova analysis confirms that awareness is significantly increases the adoption of AI tools.

Inference

Since the significant value is less than 0.05, the null hypothesis is rejected and the alternative hypothesis is accepted. Therefore, the awareness plays a significant role in adoption of AI tools among SME in Chennai City.

Hypothesis 3: Influence on the usage of AI tools and Operational efficiency

The following hypothesis is used to find the relationship between the usage of AI and the operational efficiency

H0: There is no significant difference between the influence on usage of AI tools and the operational efficiency.

H1: There is significant difference between the influence on usage of AI tools and the operational efficiency.

Table 4.3.4 Influence of the usage of AI tools and operational efficiency using regression

Anova

Model	Sum of squares	df	Mean square	F	Sig.
Regression	4.493	1	4.493	2.753	.099
Residual	323.087	198	1.632		
Total	327.580	199			
Dependent Variable: *Operational efficiency*					
Source: *Primary Data*					

Coefficients

Model	Unstandardized coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
Constant	22.014	.244		90.348	.000
AI tools	.044	.027	.117	1.659	.099
Source: *Primary Data*					

Interpretation

The results shows that the above table has no significant relationship between the usage of AI tools and its efficiency among SME food stalls in Chennai City. The regression result (Sig=.099) shows that the usage of AI tools has no operational efficiency of among SME food stalls in Chennai City.

Inference

Since the significant value is more than 0.5, the null hypothesis is accepted and the alternative hypothesis is rejected. Therefore, there is no significant relationship between the influence on the usage of AI tools and the operational efficiency.

Hypothesis 4: Impact on AI adoption tools and customer satisfaction

The following hypothesis is used to find the relationship between the impact on AI adoption and customer satisfaction.

H0: There is no significant relationship between the adoption of AI tools and customer satisfaction

H04: There is significant relationship between the adoption of AI tools and customer satisfaction.

Table 4.3.5 Impact of AI adoption tools and Customer satisfaction using Regression

Anova

Model		Sum of squares	df	Mean Square	F	Sig.
1	Regression	10.364	1	10.364	.1604	.207
	Residual	1279.636	198	6.463		
	Total	1290.000	199			
Dependent Variable: *Customer Satisfaction*						
Source: *Primary Data*						

Coefficients

	Model	**Unstandardized Coefficients**		**Standardized Coefficients**	**t**	**Sig.**
		B	**Std Error**	**Beta**		
1	(Constant)	21.251	.266		79.839	.000
	AI tools usage	.070	.055	.090	1.266	.207
Dependent Variable: *Customer satisfaction*						
Source: *Primary Data*						

Interpretation

The results shows from the above table has no significant relationship between the adoption of Artificial Intelligence tools and customer satisfaction of among SME food stalls in Chennai City. The regression result (Sig=.207) shows that the adoption

AI tools does not have any significant relationship between the customer satisfaction of among SME food stalls in Chennai City.

Inference

Since the significant value is more than 0.05, the null hypothesis is accepted and the alternative hypothesis is rejected. Therefore, there is no significant relationship between the between the adoption of AI tools and customer satisfaction.

Hypothesis 5: Challenges and barriers of adopting AI

The following hypothesis is used to find the challenges and barriers of adopting Artificial Intelligence.

H0: There is no significant difference in the level of challenges faced by SME food stalls on basis of adopting AI.

H05: There is significant difference in the level of challenges faced by SME food stalls on basis of adopting AI.

Table 4.3.6 Challenges and barriers of adopting AI using ANOVA

	Sum of squares	**df**	**Mean Square**	**F**	**Sig.**
Between Goups	153.991	7	21.999	3.692	.001
Within Groups	1143.964	192	5.958		
Toal	1297.955	199			
Dependent variable: *Challenges and barriers*					
Source: *Primary Data*					

Interpretation

The results shows from the above table is a positive difference between the AI adoption and the challenges and barriers of SME food stalls in Chennai City. The Anova result (Sig= .001) shows that the challenges and barriers affects the adoption of Artificial Intelligence. Further, the Anova analysis confirms that the adoption of Artificial Intelligence is significantly influence the challenges and barriers of SME food stalls.

Inference

Since the significant value is less than 0.05, the null hypothesis is rejected and the alternative hypothesis is accepted. Therefore, the adoption of AI plays a significant role in the challenges and barriers of SME food stalls in Chennai City. There is significant difference in the level of challenges faced by SME food stalls on basis of adopting AI.

Summary

5.1 Introduction

This chapter includes the findings of this study, suggestions ,overall conclusions and recommendations for future research. The primary objective of this study to identify the level of awareness and adoption of AI based tools among SME food stall owners which influence and examine the awareness and adoption, customer engagement, operational efficiency. This study also gives the challenges and barriers faced by SME owners on adopting AI in their day to day life.

The data is collected from the 200 SME owners using SPSS test like Regression and Anova to analyse the relationship between the AI tool usage and operational performance. The results gives about the usefulness of AI tools of SME owners in Chennai City.

5.2 Summary of Findings

This is based on the data collection and analyzed in SPSS, the findings are

5.2.1 Awareness and Adoption of AI tools

The study finds that most of the SME owners in Chennai City have a medium level awareness on AI tools and technologies in their business. Most of the food stall owners are familiar with digital mobile based payments like UPI and Gpay. Only some of the few SME owners like in small restaurants know and use AI software and also in some beverages shop and Bakeries shops. In food truck and Roadside shops, the SME owners use UPI and Gpay i.e digital payments.

However, In this study a small percentage of SME stall owners like Pani puri shops still rely on traditional methods. But this is not due to lack of awareness, But this is due to lack of knowledge using technology and AI.

5.2.2 Customer Engagement

Here, In this study there is no relationship between the adoption of AI tools and satisfaction. Some shop keepers even though they use AI, the customer preferences changes quickly. In Small restaurants, the sales depends on the customer preferences, If a food is sold more which means customers preference is more on a particular menu. But human beings taste won’t be relying on the same preferences, it changes. Also it applicable same on the bakery and beverages. In big SME like Small restaurants, beverages and bakeries, AI presents but does not give satisfaction. In small SME like roadside shop and food truck, AI is absent which don’t affect the

satisfaction. Therefore customer mainly depend on the quality of food and their taste buds and not technology.

5.2.3 Operational Efficiency

Here, In this study there is no relationship between the adoption of AI and operational efficiency. Some shopkeepers even though they use AI, there is less operational efficiency. This is because AI works along with human efforts and also most of SME owners in small restaurants, beverages and bakery shops use AI for basic purpose like billing and payments activities and they use Petpooja, Loyverse and MyBilBook. But not for the entire performance. Because AI is not fully integrated into businesses in Chennai. In SME like roadside shop and food truck they use digital payments like Gpay and Phonepay which is very partly integrated with AI in Chennai City.

5.2.4 Challenges and Barriers

Here, In this study there is significant difference in the level of challenges faced by the SME food stalls on adopting AI. Most of the shopkeepers face difficult in adopting AI. This is because AI is a very new concept but suddenly spread quickly around the globe. Using digital is slightly different from using AI, But AI at present synced well with digital technologies, every digital technologies has AI. So it became quite mandatory to use AI by shopkeepers. There is more challenges and barriers faced by all SME food stall owners of small restaurants, beverages and bakeries where they use for quite purposes like Billing and payments and also they feel that AI is costly when they want to use management and automation and sometimes its complicated. The SME like food truck and roadside shop has many barriers where most of the SME owners are uneducated. They need training to use AI. Lack of knowledge and high cost are the barriers.

5.2.5 Overall Performances

Here, In this research there is no significant relationship between in the adoption of AI and the overall performances of SME food stalls in Chennai City. The SME like small restaurants, beverages and food truck, AI plays a significant role in billing and payment and sales and profit tracking rather on the management of stock, cost, customer preferences, hygiene and taking orders. They feel like AI is costly and sometimes its complicated to manage every AI. Because AI is a broad classification. In SME like food truck and road side food shops they use only digital payments like UPI and Phone pay, where they use AI very rarely or not.

5.3 Suggetsions

Based on the findings the following suggestions are proposed to improve the overall performance of SME food stalls in Chennai City.

1. To increase the awareness and adoption training programs for SME food stall owners like food truck and road side shops. Conducting workshops and seminar for promoting the usage of AI and giving suggestions about free AI to improve efficiency and overall performance.Conducting training programs for small restaurants, beverages and bakeries by promoting the AI techniques to use when they feel complicated.
2. The Government should take initiative by proving free AI tools for small shops. Providing low cost digital AI tools for small shops to use AI efficiently and productively.
3. The software developers should create simple and user-friendly AI tools for small shops where they can use them. They can also create a low cost AI tools like need less internet facility.
4. SME can also invest on managing stock, cost and customer management rather only on billing and payment to improve overall efficiency and operational performance.
5. The shopkeepers or owners should motivate employees on the usage of AI tools.

5.4 Conclusion

This study examine the usage of AI on overall performance of SME food stalls in Chennai City through awareness and adoption of AI, efficiency of AI usage tools, Customer satisfaction and the challenges and barrier faced on using AI tools. But SME like small restaurants, beverages and bakeries use AI based tools which is on the purpose of billing and payments, sales and profit tracking rather on managing stock, cost and customer management. There is lack of training on using these softwares. They use softwares like Petpooja, Loyverse and MyBillBook. Whereas, In SME like food truck and roadside shop they use digital payments like Phone Pay and Gpay which is partly AI. Due to their educational background and lack of knowledge they feel hesitate to use AI.

Overall, the study gives the conclusion of that AI adoption is slowly increasing among SME food stalls in Chennai City. There should be a improvement on training, accessibility and usage of AI.

5.5 Scope for Future research

- The present study focuses on SME food stalls in Chennai City. There are some areas are need to be covered in Chennai City.
- The present study focuses on the usage of AI activities such as Billing and Payment and sales and profit tracking. There are some areas like prediction of sales and profit and automated AI tools are need to be covered.
- The present study focuses on the SME which is related to only food stalls. There are some sectors like retail stores, manufacturing sectors and also service sectors.
- Moreover, the further research can also include role of Government policies, digital infrastructure and financial support programs in encouraging the adoption of AI. As AI is ruling the entire world.

APPENDIX

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6.2 Questionnaire

Part A – Demographic Questions:

1. Gender

☐ Male

☐ Female

2. Age Group

☐ Below 25

☐ 25–35

☐ 36–45

☐ 46–55

☐ Above 55

3. Educational Qualification

☐ School Level

☐ Diploma

☐ Undergraduate

☐ Postgraduate

4. Type of Food Stall

☐ Street Food / Roadside stall

☐Small Restaurant / Food court outlet

☐Bakery / Snack shop

☐Food Truck

☐Beverage-focused stall

5. Nature of ownership

☐Sole Proprietorship

☐Partnership

☐ Family Business

6. Years of Business Experience

☐ Below 3 years

☐ 3–5 years

☐ 6–10 years

☐ Above 10 years

7. Number of Employees

☐ Below 5

☐ 5–10

☐ Above 10

8. Average Daily Sales

☐ Below 5,000

☐ 5,001–10,000

☐ 10,001–20,000

☐ Above 20,000

9. Which of the following devices do you currently use in your food stall? (Select all that apply)

☐ Smartphone (Android / iPhone)

☐ Desktop computer / Laptop

☐ Ordering Screen

☐ Others

☐ Ordering screen

☐ None of the above

10. In which of the following activities do you use digital apps or software in your food stall? (Select all that apply)

☐ Billing and payment collection

☐ Managing stock / ingredients

☐ Taking orders (dine-in / takeaway / delivery)

☐ Managing online food delivery orders (Swiggy / Zomato, etc.)

☐ Customer communication (offers, feedback, messages)

☐ Sales or profit tracking

☐ Digital payments

☐ I do not use any digital app or software

11. Which of the following digital tools or apps do you currently use for billing or order management in your food stall? (Select all that apply)

☐ Petpooja

☐ Go Frugal

☐ WhatsApp Business

☐ Loyverse

☐ Swiggy

☐ MyBillBook

☐ Phone pay/Gpay

☐ Other billing / POS app (please specify): _____

☐ I do not use any billing or POS app

12. Section B: Awareness and Adoption of AI

Likert scale questions 1-Strongly Disagree, 2-Disagree ,3-Neutral ,4-Agree ,5-Strongly agree

S.no	Awareness and Adoption of AI	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
1	I am aware of the different AI tools and applications available for food stalls (e.g., POS, billing apps, order management apps).					
2	I understand how AI tools can improve the daily operations of my food stall, such as billing, inventory management, and order					

	processing.					
3	I am familiar with specific AI-enabled tools such as Petpooja, Go Frugal, Posit, Ezee, Vyapar, MyBillBook, or Zoho Invoice.					
4	I actively use, or have experience using, one or more AI-enabled tools in my food stall.					
5	I make an effort to stay updated about new AI technologies and tools for food stalls through online resources, workshops, or training programs.					

13. Section C: AI and Operational Efficiency

S.no	AI and Operational Efficiency	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
1	I am aware that AI-based tools can help improve the speed of service in my food stall.					
2	I understand that AI applications can assist in better inventory management and stock control.					
3	I am aware that AI-enabled billing systems can reduce errors and save time.					
4	I understand that AI tools have the potential to reduce food wastage and optimize resource usage.					

5	I believe that adopting AI can improve the overall operational efficiency of my food stall.					

14. Section D: AI and Customer Satisfaction

S.no	AI and Customer Satisfaction	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
1	AI has improved customer convenience in my food stall through faster billing, digital menus or online ordering.					
2	AI has helped me understand customer preferences such as popular food items and peak demand times.					
3	AI has improved overall customer satisfaction in my food stall.					
4	AI has influenced customers to prefer food stalls that use digital or AI-enabled services.					
5	AI has contributed to repeat purchases and customer loyalty in my food stall.					

15. Section E: Impact on Overall Performance

S.no	Impact on Overall Performance	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
1	I am aware that the adoption of AI-based or digital tools can help increase sales in my food stall.					
2	I understand that AI-based tools can help reduce					

	operational costs such as wastage, errors or manual effort.					
3	I am aware that AI tools can support better decision-making related to pricing, purchasing, and stock management.					
4	I feel that using AI digital tools gives more benefits than the money I spend on them.					
5	I recognize that AI can play an important role in improving the overall performance of my food stall.					

16. Section F: Challenges and Barriers in AI Adoption

S.no	Challenges and Barriers in AI Adoption	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
1	I am aware that the high cost of AI-based or digital tools is a major challenge for food stalls..					
2	I understand that a lack of technical knowledge makes it difficult to use AI or digital tools effectively.					
3	I recognize that limited training or guidance creates difficulties in adopting AI-based tools.					
4	I am aware that poor internet connectivity or digital infrastructure affects the use of AI systems.					
5	I believe that fear of technology complexity discourages food stall owners from adopting AI tools.					

17. Any suggestions to improve AI adoption for sustainability among SME food stalls in Chennai City?

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24pcm16 Jeshwin

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